Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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Abstract

Large language models are increasingly deployed in decisions that turn on moral judgement, from content moderation to clinical prioritisation, yet they answer as if the whole world thinks like the West. The Moral Machine experiment [Awad et al., 2018] showed this is wrong at scale: 40 million judgments across 233 countries reveal that moral preferences are systematically structured by culture, and a model that ignores this variation does not merely underperform. It imposes one society's intuitions on all others. Existing fixes do not scale to global deployment: fine-tuning needs per-country preference data and GPU budgets, reward-guided decoding needs per-country reward models, and activation steering needs access to model internals that black-box APIs do not expose. We work in the only regime that scales: inference time, with no weight updates, no training data, and no internal access. The key observation is that within-country demographic *disagreement*, not consensus, is the steering signal: when culturally grounded personas agree, the base model is already calibrated; when they disagree, the spread tells us what to fix and by how much. DISCA (Disagreement-Informed Steering for Cultural Alignment) instantiates each country as a panel of four World-Values-Survey-grounded persona agents [Haerpfer et al., 2020], converts their disagreement into a bounded, loss-averse correction whose magnitude is set by the panel's variance, and shrinks the correction toward zero when the estimate is unreliable. Across 20 countries and 7 open-weight backbones (2B–70B) from five model families, DISCA reduces cultural misalignment on MultiTP [Jin et al., 2025] by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios; a 14B backbone with DISCA reaches lower absolute misalignment than a vanilla 70B model.

1 Introduction

Large language models increasingly mediate decisions that turn on moral judgement: content moderation policies, clinical prioritisation guidelines, autonomous-vehicle behaviour specifications, and high-stakes recommendation pipelines. In each of these settings, the model's implicit moral preferences become the system's moral preferences. Yet a growing body of evidence shows that these implicit preferences are not culturally neutral. The Moral Machine experiment collected 40 million judgments across 233 countries and showed that moral preferences are systematically structured by culture [Awad et al., 2018]; large language model (LLM) outputs correlate most strongly with respondents from Western, educated, industrialised, rich, and democratic populations [Henrich et al., 2010, Santurkar et al., 2023]; and post-training alignment introduces further unintended cultural shifts [Ryan et al., 2024, Zewail et al., 2026]. A model that ignores this variation does not merely underperform on a benchmark; it imposes one society's intuitions on every user it serves.

The cultural gap is now well-documented at the model level. The MultiTP benchmark [Jin et al., 2025] extends the Moral Machine protocol to 107 languages with country-specific Average Marginal

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Component Effects (AMCEs), and reports that none of 19 evaluated LLMs recovers the country-level structure of human moral preferences. The asymmetries are substantial: Japan ranks first among countries on preference for sparing pedestrians, China ranks 116th, and current models do not capture either extreme. Concurrent work shows that sociodemographic persona prompts shift LLM moral decisions two to four times more than they shift human decisions [Kim et al., 2025], indicating that the models are not insensitive to cultural framing, but that the framing they receive at deployment time does not match the cultural distribution they are asked to serve.

Closing this gap is harder than it looks. Three families of methods have been proposed, and each demands resources that do not scale to the long tail of countries served by a single deployed model. Fine-tuning and culture-aware adapters require per-country preference datasets and GPU budgets for every target population [Zhang et al., 2026, Yao et al., 2025]; reward-guided decoding requires a separate trained reward model per country [Khanov et al., 2024, Mudgal et al., 2024]; and activation steering requires write access to model internals [Arditi et al., 2025, Zou et al., 2023, Turner et al., 2024], which black-box APIs do not expose. Any deployable solution must therefore work at inference time, must not assume per-country reward models or labelled preference data, and must not assume access to model internals beyond what an API exposes. We refer to this as the black-box, public-data-only regime, and argue that it is the only regime in which cultural alignment can scale from a handful of curated countries to the hundreds of cultures a globally-deployed model actually serves.

We make the following observation. When certain culturally grounded persona prompts representing the same country face the same moral dilemma, the dispersion of their decoding-time logit gaps is itself informative. If they agree, the base model is already calibrated for that country and no correction is needed; if they disagree, the spread of their gaps encodes both the direction and the magnitude of the needed correction. Our method, DISCA (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2020], evaluates them in a single batched forward pass, and converts the panel's variance into a bounded, loss-averse logit correction whose magnitude is automatically attenuated when the underlying estimate is unreliable. We show that this attenuation is not heuristic: it implements a variance-aware shrinkage estimator whose shrinkage weight is determined by the panel's empirical disagreement, formally characterising when the correction can be trusted.

Across 20 countries spanning four continents and seven open-weight backbones (2B–70B parameters) from five model families, DISCA reduces cultural misalignment on MultiTP by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios. The strongest absolute alignment is achieved by Phi-4 (14B), which surpasses a vanilla Llama-3.3-70B backbone despite using one fifth of the parameters; this suggests that, in the regime we study, calibration competes with scale rather than merely complementing it. An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] reveals a clean scope boundary: DISCA steers values, where the decision reduces to a scalar logit gap, but does not transfer to factual recall, where the decision is a single token in a large vocabulary. Failure modes are architecturally rather than culturally determined and are diagnosable from the base model's vanilla logit conditioning alone. Primary metrics (MIS and diagnostics) are defined in §4.1.

Contributions.

- An inference-time cultural alignment method that requires no weight updates, no per-country reward models, and no white-box access.
- A formal characterisation of disagreement-driven shrinkage. To our knowledge, the first inference-time alignment method to use independent-run disagreement as a continuous reliability signal.
- An empirical evaluation across 20 countries, four continents, seven open-weight backbones, and three evaluation formats (binary moral dilemmas, open-ended scenarios, factual QA), demonstrating that a 14B model with the proposed recipe matches or surpasses a 70B model without it.

2 Related Work

Benchmarks and cultural evaluation. The Moral Machine experiment [Awad et al., 2018] established that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs. Takemoto [2024] and Ahmad and Takemoto [2025] showed that LLM alignment varies widely across model families. Kim et al. [2025] demonstrated that sociodemographic personas induce moral-decision shifts in LLMs 2–4$\times$ larger than in humans, with political orientation producing the largest effect (a "partisan sorting" phenomenon absent in human respondents). Their finding that simple persona prompts amplify rather than correct misalignment directly motivates DISCA's approach: we use WVS-grounded personas not to steer decisions toward a single viewpoint but to measure within-country disagreement and convert it into a bounded correction. Existing moral benchmarks remain English-centric [Hendrycks et al., 2021, Jin et al., 2022, Nie et al., 2023] or cover fewer languages [Kirk et al., 2024, Durmus et al., 2023, Qiu et al., 2025]. We validate on MultiTP's binary trolley vignettes, the BLEnD factual cultural QA benchmark [Myung et al., 2024] (Appendix A7), and a dedicated open-ended moral-dilemma track (§4.4). On the diagnostic side, LLM outputs correlate most with WEIRD respondents [Santurkar et al., 2023], RLHF introduces unintended cultural shifts [Ryan et al., 2024], and recent work shows LLMs systematically stereotype non-Western moral values [Zewail et al., 2026].

Inference-time alignment. The pluralistic alignment vision [Sorensen et al., 2024] argues that inference-time methods are a natural vehicle for cultural adaptation, but existing approaches each sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs per-country reward models; culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-culture training. Recent logit-level steering [An et al., 2026, Khanuja et al., 2026] demonstrates that the logit surface is a richer control interface than appreciated, but neither targets country-conditioned moral calibration. Black-box alignment modules [Bobbili et al., 2025, Chen et al., 2025] still require a training stage, and adaptive IS in pre-logit space [Kanai et al., 2025] requires activation access. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] applies the same PT value shape we use, but at training time; DISCA applies it at *test time*, per scenario, aggregating heterogeneous personas rather than shaping a trained policy.

Distinction from superficially similar paradigms. DISCA may superficially resemble ensemble methods, but the resemblance is architectural, not algorithmic. In self-consistency [Wang et al., 2023] and LLM debate [Du et al., 2024, Liang et al., 2024], diversity is noise to be eliminated via majority vote; in standard ensembles, multiple outputs are averaged to reduce variance of the *prediction*. In DISCA, diversity *is* the signal: the *spread* of persona logit gaps, not the mode, determines the correction magnitude. This distinction is not merely conceptual. We prove (Proposition 1) that the disagreement-driven correction is a James–Stein shrinkage estimator [James and Stein, 1961] that strictly dominates unshrunk consensus in worst-case MSE: ensembles average to reduce output variance, whereas DISCA uses within-panel variance as a sufficient statistic to *control* correction magnitude. Standard calibration (temperature scaling, MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]) adjusts confidence isotropically; DISCA applies anisotropic, loss-averse corrections. Table 4 quantifies the gap versus removing personas or the reliability gate.

To our knowledge, DISCA is the first training-free method targeting country-level moral calibration. The method is grounded in control-as-inference, Prospect Theory, and importance-sampling variance control (Appendix A13).

3 Method: DISCA

DISCA aligns the model by creating an adjustment derived from the agents' consensus, and the agreement among the agents controls how much of it actually applies. The magnitude of the adjustment is not fixed: when the four agents agree on what the country's preference should be, we apply the adjustment in full; when they disagree, we apply only a fraction of it, scaled by how much the agents disagree. This inverts the standard combination structure used by self-consistency [Wang et al., 2023], multi-agent debate [Du et al., 2024, Liang et al., 2024], and ensembles, where N candidate outputs are aggregated by majority or averaging and disagreement among them is treated as

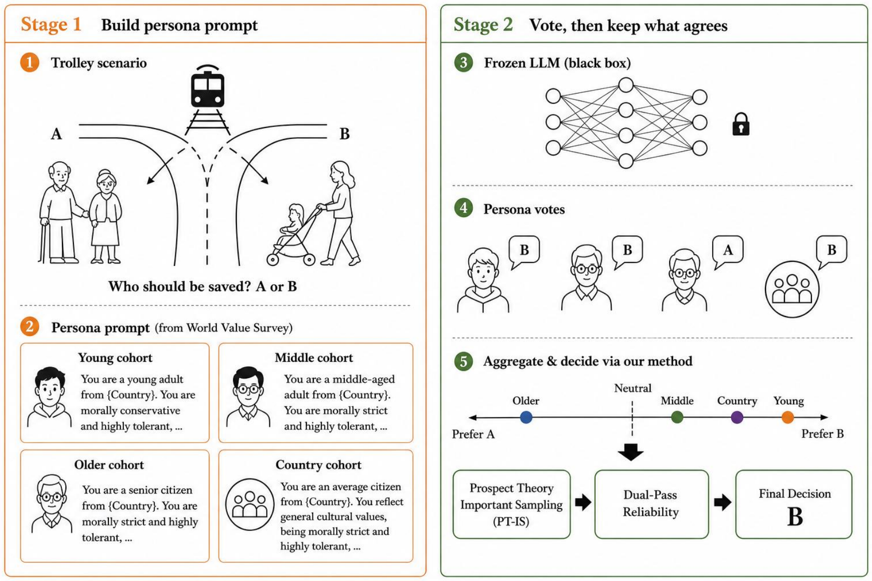

Figure 1: **DISCA overview.** Stage 1 builds WVS-grounded persona prompts for a trolley scenario in country c; Stage 2 runs a frozen large language model (LLM) on the base prompt and each persona, aggregates persona-level signals in logit space, and applies *Prospect-Theory importance sampling* (PT–IS; §3.3) together with a dual-pass reliability gate to obtain the final sparing probability. Pseudocode and the six MultiTP attribute–temperature pairs appear in Appendix A1.

noise to be suppressed. DISCA instead treats disagreement as the quantity that controls how much adjustment to apply.

3.1 Setup

We consider a frozen language model f_θ on a forced-choice task: each input x asks the model to choose between two options via decision tokens A and B. Let $z(x) = [z_a, z_b]$ denote the model's logits at the decision-token position, and let $\delta(x) = z_b - z_a$ be the resulting decision gap. For target country c, we assume access only to f_θ as black-box and a public characterisation of country c's population (in our case, World Values Survey microdata [Haerpfer et al., 2020]); we do not assume per-country preference labels, reward models, or write access to θ. Our goal is to compute, from f_θ alone, a per-scenario logit adjustment $\delta^\star(x, c)$ such that the adjusted gap $\delta(x) + \delta^\star(x, c)$ produces decisions better aligned with country c's preferences. Alignment is measured against a country-level reference vector; the benchmark and metric are described in 4.1. The remainder of Section 3 focuses on how $\delta^\star$ should depend on the disagreement, what formal property this dependence has, and how to estimate the correction.

3.2 The disagreement principle

Suppose we evaluate the same scenario under N culturally grounded prompts representing the same target country, each producing a logit gap δ_i. Each δ_i is a noisy estimate of what country c's population-level preference would imply for this scenario. If the δ_i cluster tightly, the four estimates converge on the same answer and the consensus is a reliable signal. If they are dispersed, the consensus is averaging over disagreeing estimates and is unreliable as a steering target. The principle is that within-agent dispersion, not consensus magnitude, controls how much of the consensus adjustment we should actually apply.

Let $\delta_1, \ldots, \delta_N$ be the logit gaps produced by N agents on a single scenario, and let δ_h denote the unobserved logit gap consistent with country c's population-level preference on that scenario. We model the agents as

$$\delta_i = \delta_h + \eta_i, \qquad \mathbb{E}[\eta_i] = 0, \qquad \mathrm{Var}(\eta_i) = \tau^2, \tag{1}$$

with the η_i assumed independent across agents and identically distributed with common variance τ^2. This is a standard hierarchical setup: δ_h is a fixed-but-unknown population parameter, and each agent is a noisy view of it. The i.i.d. structure is a modelling simplification, since all agents share base model f_θ and the scenario.

Define the consensus $\bar{\delta} = \frac{1}{N}\sum_i \delta_i$, the within-agent variance $D^2 = \frac{1}{N-1}\sum_i(\delta_i - \bar{\delta})^2$, the consensus adjustment $\Delta = \bar{\delta} - \delta_{\text{base}}$, and the oracle adjustment $\Delta_h = \delta_h - \delta_{\text{base}}$. Where δ_{base} is the logit gap produced by the unconditioned base prompt (without persona or country-related prompts) on the same scenario.

We will apply a scaled version of Δ as our adjustment: $\hat{\Delta}(\gamma) = \gamma \cdot \Delta$ for some $\gamma \in [0, 1]$. We choose γ by minimising the mean squared error against the unobserved oracle,

$$\text{MSE}(\gamma) \;=\; \mathbb{E}\big[\big(\hat{\Delta}(\gamma) - \Delta_h\big)^2\big] \tag{2}$$

Proposition 1 (Variance-aware shrinkage)**.** *Under the model in* (1)*:*

(i) *D^2 is unbiased for τ^2, and the variance of the consensus adjustment is* $\text{Var}(\Delta) = \tau^2/N$.

(ii) *Among shrinkage estimators of the form $\hat{\Delta}(\gamma) = \gamma \cdot \Delta$ with $\gamma \in [0, 1]$, the MSE-minimising shrinkage factor is*

$$\gamma^\star = \frac{\Delta_h^2}{\Delta_h^2 + \tau^2/N}, \tag{3}$$

which is bounded in $(0, 1)$ and monotone-decreasing in τ^2.

The proof is in App. A2. The Proposition's two parts work together to motivate the method. Part (ii) gives the optimal $\gamma^\star$ in closed form, but it is not directly computable: $\gamma^\star$ depends on τ^2 and on Δ_h, neither of which we observe. Part (i) gives us a way to estimate τ^2 from data we already have: the within-agent variance D^2 has expected value equal to τ^2. But Δ_h has no analogous estimator from the agents' outputs, so we cannot compute $\gamma^\star$ exactly. Instead, in 3.3 we construct an estimator that depends on D^2 and matches the shape of $\gamma^\star$: bounded in $(0, 1)$, monotone-decreasing in the variance estimate.

3.3 The DISCA estimator

Proposition 1 tells us what shape the shrinkage should have but not how to compute it: $\gamma^\star$ depends on the unobservable Δ_h. We now construct an estimator $\delta^\star$ that is computable from the four agents' outputs alone and matches the qualitative shape of $\gamma^\star \cdot \Delta$ — bounded, monotone-decreasing in the variance estimate, and smooth.

We use N=4 agents per target country: three age cohorts (young, middle-aged, older) plus a country-wide aggregate, each instantiated as a native-language system prompt built from cohort- and country-level aggregates of the WVS cultural profile of Greco et al. [2026] (full prompts in App. A15). To remove the order bias of forced-choice decoding, we evaluate every scenario under both (A, B) and (B, A) orderings and take the order-symmetrised gap $\delta_i = \frac{1}{2}(\delta_i^{(AB)} - \delta_i^{(BA)})$ as the agent's output, which keeps the η_i in (1) mean-zero.

$$\bar{\delta} = \frac{1}{N}\sum_{i=1}^{N}\delta_i, \tag{4}$$

$$r_i = \delta_i - \delta_{\text{base}}. \tag{5}$$

3.3.1 Loss-averse importance sampling

The plain consensus adjustment $\Delta = \bar{\delta} - \delta_{\text{base}}$ moves the base model toward the average direction because $\bar{\delta}$ is the average consensus. The problem is that an average direction can hide a cohort that strongly disagrees: three cohorts mildly in favour and one cohort strongly against will average out to "mildly in favour," and the consensus will move the model in a direction that one cohort actively rejects. We want a steering rule that avoids this. One that prefers a direction nobody strongly dislikes over a direction with a higher average but a strong dissenter.

We frame this as a cooperative-bargaining problem among the cohorts [**??**]. Each cohort is a player with its own utility over correction directions, each is loss-averse which mean a small misalignment with one's preferred direction is felt more than a same-sized improvement, and we want to pick a direction that all players can collectively accept. This framing organises three design choices: where to look for candidate, how each player scores a candidate, and how the all scores combine into a group verdict.

Rather than committing to Δ, we draw K candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ in a neighbourhood of the consensus. Because the consensus gives us only one direction; to find a direction that all cohorts can live with, we have to look at neighbouring directions too. So each ϵ_k is a candidate: an alternative correction the players will be asked to evaluate. The per-cohort gain $g_{i,k}$ records how much ϵ_k improves cohort i's alignment relative to the base prompt; positive $g_{i,k}$ means the candidate moves the model toward cohort i's preferred direction, negative means it moves the model away.

$$g_{i,k} = -\left|(\delta_i - \delta_{\text{base}}) - \epsilon_k\right| + \left|\delta_i - \delta_{\text{base}}\right|. \tag{6}$$

We need a scoring rule $v(\cdot)$ that is concave on the gain side so additional gain beyond what's already aligned is worth less and steeper on the loss side so any cohort loss matters more than the same-size gain elsewhere. The canonical such rule in the decision-theory literature is the Kahneman–Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992]:

$$v(z) \;=\; \begin{cases} z^{\alpha} & z \geq 0, \\ -\kappa\,(-z)^{\alpha} & z < 0, \end{cases} \tag{7}$$

with curvature $\alpha \in (0, 1]$ and loss-aversion $\kappa \geq 1$. The asymmetric kernel v is a controller-design choice: it gives us an aggregator that penalises misalignment more than it rewards incremental alignment.

Cooperative bargaining sits between two extremes: a purely *utilitarian* verdict averages the players' utilities and ignores how the gains are distributed, while a purely *collective* verdict asks how the deal serves the group as a whole. We use a blend of the two:

$$U_{\text{total}}(\epsilon_k) \;=\; (1 - \lambda_{\text{coop}})\,\frac{1}{N}\sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \;+\; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{8}$$

where $g_{\text{cons},k}$ measures the deal's gain against the consensus direction $\bar{\delta}$ rather than against any single cohort. The first term is the utilitarian verdict over loss-averse players; the second is the collective verdict scored by the same loss-averse rule. The coefficient $\lambda_{\text{coop}} \in [0, 1]$ controls the bargaining position, with $\lambda_{\text{coop}} = 0$ purely utilitarian and $\lambda_{\text{coop}} = 1$ purely collective. We use $\lambda_{\text{coop}} = 0.7$, on the collective side; the sensitivity analysis is in App. A14. The division by σ inside v scores each gain in units of the perturbation scale, so the search radius σ and the value-function shape (α, κ) can be set independently.

The natural choice, returning the highest-utility candidate, is unsafe here. The utility $U_{\text{total}}(\epsilon_k)$ is itself a noisy quantity. It depends on the random draw ϵ_k and on the agent gains $g_{i,k}$, both of which carry stochastic error. Picking the argmax over K noisy scores selects the candidate that drew favourable noise on this sample, not the candidate with the highest true utility. We therefore aggregate all candidates with weights proportional to their utility:

$$\delta_{PT-IS} \;=\; \frac{\sum_{k=1}^{K} w_k\, \epsilon_k}{\sum_{k=1}^{K} w_k}, \qquad w_k \propto \exp(U_{\text{total}}(\epsilon_k)). \tag{9}$$

This form is not ad hoc: it is the path-integral solution to the stochastic optimal control problem of selecting a correction direction under a noisy utility function [Williams et al., 2018, Levine, 2018]. The high-utility candidate dominates the average, but near-winners contribute as well, and stochastic error in any one candidate's utility is suppressed by the contributions of the others. We refer to the full procedure as *loss-averse importance sampling*, or PT-IS for short.

3.3.2 Dual-pass reliability gate

PT-IS gives us a single correction direction $\delta^{\star}_{\text{IS}}$ per scenario, but this point estimate hides a finite-sample problem. The aggregation in (9) averages over K stochastic candidates, and the result depends on which candidates happened to be drawn. On scenarios where the cohorts agree, the candidates score similarly and any reasonable sample gives a similar $\delta^{\star}_{\text{IS}}$. On scenarios where the cohorts disagree, the utility surface is uneven and the same K-sample budget can produce meaningfully different aggregates on two independent draws. From a single $\delta^\star_{\rm IS}$ the caller cannot tell which regime they are in.

§3.2 told us what to do in this situation. Proposition 1(ii) said that the optimal correction shrinks toward zero in proportion to the estimator's uncertainty: full strength when the estimate is reliable, zero when it is not. We need to detect the unreliable case and contract the correction accordingly. Doing this requires two things — a way to measure the unreliability of $\delta^\star_{\rm IS}$ on a given scenario, and a rule for turning that measurement into a contraction factor.

Measuring the unreliability. We split the candidate budget in half and run PT-IS twice on disjoint sub-budgets, producing two independent corrections $\delta^\star_1$ and $\delta^\star_2$ on the same scenario. If the two replicates land close to each other, the underlying utility surface is well-behaved and $\delta^\star_{\rm IS}$ is trustworthy. If they land far apart, the aggregator was genuinely sensitive to which candidates got drawn, and $\delta^\star_{\rm IS}$ is finite-sample noise as much as signal. The squared inter-replicate gap

$$V_r \;=\; (\delta^\star_1 - \delta^\star_2)^2$$

is the operational analogue of the within-cohort variance D^2 from Proposition 1(i): both measure disagreement among same-source quantities — D^2 across cohorts, V_r across replicates of the aggregator — and both are computable from quantities the four agents already produced.

Contracting the correction. We use V_r to multiplicatively shrink the correction:

$$r \;=\; \exp\!\big(-V_r/s\big) \;=\; \exp\!\big(-(\delta^\star_1 - \delta^\star_2)^2/s\big) \;\in\; (0,1], \tag{10}$$

with bandwidth $s > 0$ controlling how fast r decays in V_r. The exponential is the simplest function with the three shape properties Proposition 1 requires: it is bounded in $(0, 1]$, monotone-decreasing in the variance proxy, and smooth. When the two replicates agree, $V_r \approx 0$ and $r \approx 1$, applying the full PT-IS correction. When they disagree, r decays toward zero without any hard threshold, and the correction contracts toward the no-op direction. The final DISCA correction is the gated average of the two replicates:

$$\boxed{\delta^\star \;=\; r \cdot \frac{\delta^\star_1 + \delta^\star_2}{2}.} \tag{11}$$

For cross-reference compatibility with earlier drafts, we denote the same reliability-gated correction as

$$\delta^\star_{\rm IS} \;=\; \exp\!\big(-(\delta^\star_1 - \delta^\star_2)^2/s\big) \cdot \frac{\delta^\star_1 + \delta^\star_2}{2}. \tag{12}$$

The three shape properties. The estimator $\delta^\star$ matches the qualitative shape of the optimal $\gamma^\star \cdot \Delta$ from Proposition 1: *(a)* the gate $r \in (0, 1]$ keeps $|\delta^\star|$ bounded by the average replicate magnitude $|\frac{1}{2}(\delta^\star_1 + \delta^\star_2)|$; *(b)* r is monotone-decreasing in the inter-replicate disagreement, mirroring the monotonicity of $\gamma^\star$ in τ^2; and *(c)* both the gate and the PT-IS weights vary smoothly with their arguments, so $\delta^\star$ has no hard-threshold discontinuities.

3.4 Extension to Open-Ended Ethical Scenarios

To extend DISCA to open-ended scenarios, the model generates free-form responses under both the original prompt and four persona-specific prompts. A separate large language model (LLM) judge then parses these outputs to return a {choice, confidence} pair. The continuous decision signal is represented by a pseudo-logit gap (the A/B logit distance) extracted by that judge. The dual-pass Prospect-Theory importance sampling (PT–IS) mechanism is applied similarly to the binary case.

4 Experimental Results

4.1 Setup

We test DISCA on MultiTP [Jin et al., 2025], the multilingual extension of the Moral Machine trolley problem [Awad et al., 2018]. Each scenario shows the model two groups of people that an autonomous vehicle could spare, and the model has to choose one of them by emitting a single decision token, A or B. Every scenario isolates one of six moral attributes (species, gender, age, fitness, social value, and the number of lives saved) while matching the two groups on everything else. An age scenario, for example, pits a young group against an older one and holds the rest of the description constant; the rate at which the model spares the younger group, averaged across many such scenarios, is then read off as its preference on age. Following the Moral Machine convention we always assign decision token B to the side the AMCE measures toward; this is a measurement convention from conjoint analysis that fixes the sign of the preference, not a normative claim about which side ought to be saved. The frozen model emits logits $z(x) = [z_a, z_b]$ at the decision position, the decision gap is $\delta(x) = z_b - z_a$, and the sparing probability under decoding temperature T_{dec} is $p_{\text{spare}}(x) = \sigma(\delta(x)/T_{\text{dec}})$. Averaging the sparing probability across the scenarios for one attribute and one country gives that country's Average Marginal Component Effect (AMCE):

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x). \tag{13}$$

Stacking the six attributes gives a six-dimensional vector $\hat{m}^{(c)}$ that we compare to the human AMCE vector $h^{(c)}$ aggregated from real Moral Machine judgments [Awad et al., 2018, Jin et al., 2025]. Our headline metric is the *misalignment score* $\mathrm{MIS}(c) = \|\hat{m}^{(c)} - h^{(c)}\|_2$, which is just the straight-line distance between the model's preference vector and the country's human one; lower is better. We report it macro-averaged over a 20-country panel that spans four continents. The country list, the preprocessing pipeline (deduplication, a quality filter, per-attribute caps, oversampling, and deterministic shuffling), the per-country slice sizes, and a cap-disabled sensitivity rerun all live in Appendices A17 and A12. Every scenario is presented in the country's native language; only the decision tokens A and B stay in English so that token-ID extraction is consistent across model families. We also report Pearson r for shape agreement and Jensen–Shannon divergence (JSD) as secondary diagnostics.

We tested 28 model–method combinations across 12 architectures from 270M to 70B parameters, with the full landscape in Appendix A6, and report seven open-weight backbones with the most robust gains: Llama-3.3-70B, Magistral-24B, Phi-4 (14B), Qwen3-VL-8B, Qwen2.5-7B, Phi-3.5-mini (3.8B), and Gemma-4-E2B (2B). The few models that degrade share a single pattern: their vanilla MIS is already low enough that there is no headroom left for any correction (we return to this in §4.5). The released hyperparameter configuration was picked from 25 variants on a small three-backbone, five-country prototyping panel (Appendix A21); how we elicit the A and B tokens consistently across model families is described in Appendix A18.

We compare DISCA against two tiers of baselines on the same 20-country Phi-4 grid (Table 1; full implementations in Appendix A19). The first tier is training-free and never sees the human AMCE: vanilla decoding, a WVS Profile Prompt that summarises the country's WVS values, a PRISM-style cultural framing prompt [Kirk et al., 2024], a fixed logit offset, activation steering [Arditi et al., 2025, Zou et al., 2023], and MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]. We exclude reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] because it assumes per-country reward models, which simply do not exist for the long tail of countries we care about. The second tier is the opposite extreme: oracle baselines that fit a single scalar (per-country temperature or margin) directly on the human AMCE [Guo et al., 2017]. They are an upper bound on what any global scalar fit can do.

4.2 Main results

A single decoding-time intervention, with no weight changes and no per-country labels, materially closes the gap between large language models and human moral preferences. Across seven backbones from five model families, DISCA reduces macro MIS by roughly ten to twenty-four percent relative to vanilla decoding (Table 2), and the two strongest models improve on every one of the twenty countries we test. A 2D PCA of the AMCE vectors tells the same story visually (Figure 3, Appendix A5): every country point migrates from the vanilla cloud toward the human cloud, none in the wrong direction. No simpler intervention comes close. On the same Phi-4 grid (Table 1) DISCA outperforms every training-free baseline by a clear margin, and even the two oracle baselines that get to peek at the human AMCE during calibration end up worse than DISCA, because fitting one global scalar overshoots the macro mean instead of recovering each country's individual shape.

Table 1: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even so they fail to match DISCA.

Method	AMCE?	MIS ↓
DISCA (ours)	No	**0.346**
PRISM-Style Prompt [Kirk et al., 2024]	No	0.384
MC-Dropout [Gal and Ghahramani, 2016]	No	0.403
Activation steering [Arditi et al., 2025]	No	0.430
Fixed logit offset	No	0.439
WVS Profile Prompt	No	0.453
Vanilla decoding	No	0.454
Margin scaling [Guo et al., 2017]	Yes	0.506
Temp. scaling [Guo et al., 2017]	Yes	0.513

Table 2: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance ($\pm$ std over three seeds $\{42, 101, 2026\}$ chosen under a fixed compute budget; per-cell std median 0.008, full multi-seed analysis in §A8.1); "Win" = countries with lower DISCA MIS than vanilla.

Model	MIS ↓	Gain (%)	Win/20	r
Llama-3.3-70B	$.668{\pm}.006$	+21.3$\pm$0.8	**20**	-0.01
Magistral-24B	$.350{\pm}.005$	+13.0$\pm$0.7	16	+0.62
Phi-4 (14B)	$\mathbf{.346}{\pm}.004$	**+23.6**$\pm$0.6	18	+0.56
Qwen3-VL-8B	$.466{\pm}.006$	+18.8$\pm$0.9	**20**	+0.22
Qwen2.5-7B	$.362{\pm}.005$	+20.0$\pm$0.8	17	+0.45
Phi-3.5-mini	$.571{\pm}.006$	+10.3$\pm$1.0	17	-0.35
Gemma-4-E2B (2B)	.479	+3.4	14	+0.46

Table 3: Open-ended track summary (20 countries, 310 scenarios each). **VAN** = vanilla decoding (no DISCA). **DISCA** = full pipeline. **MIS** = misalignment score (§4.1). $\Delta\%$ reports mean relative MIS reduction across four backbones; per-country breakdown in Appendix A3.1.

Model	**Vanilla**	**DISCA**	**Δ%**
Llama-3.3-70B	0.471	0.439	+6.85%
Phi-4 (14B)	0.324	0.302	+6.67%
Qwen2.5-7B	0.321	0.306	+4.55%
Phi-3.5-mini	0.521	0.510	+2.13%

The per-country breakdown (Table 6, Appendix A3) confirms that the macro gains are not concentrated in a handful of lucky cells. Geographic patterns, large single-country swings, and the harder regions (such as those with sparser WVS coverage) are all detailed there. To rule out artefacts, we also ran nine independent robustness checks and three random seeds; every conclusion sits comfortably within the bootstrap noise floor (Appendices A11 and A8.1).

4.3 Calibration competes with scale

The single most striking finding is a crossing in the curve of MIS against model size (Figure 4, Appendix A5): a 14B model with DISCA (Phi-4) reaches lower absolute misalignment than a 70B model without it (Llama-3.3-70B). Per-dimension error analysis (Table 9, Appendix A4) explains why. Different models hit a wall on different attributes. Weaker models are dominated by Utilitarianism errors that are simply too large for any persona-based correction to close. Better-calibrated models have already resolved Utilitarianism and Species, so their bottleneck shifts to Social Value and Age, two attributes that DISCA's importance-sampling stage actually moves the needle on. Phi-4 turns out to be the most balanced of the seven across all six dimensions, and it is that balance, rather than raw parameter count, that lets it lead the panel despite being five times smaller than the 70B baseline.

4.4 Generalisation beyond binary dilemmas

The same disagreement signal also helps in open-ended ethical generation (§3.4), not just in forced binary choices. All four backbones we tested in this regime improve on average, with gains up to roughly seven percent (Table 3). The few small negative shifts are not bugs but the safety gate doing its job: when the dual-pass reliability check sees the two replicates disagree, it suppresses the correction rather than commit to a noisy one. A logit-noise stress test (Appendix A3.2) shows that the full pipeline keeps a ten to twelve percent MIS advantage over an ungated variant—the same PT–IS stack with the agreement-based shrinkage of Eq. 12 disabled—across noise levels $\sigma \in [0, 2]$, and the gap actually widens as the noise grows.

Table 4: Cross-backbone ablation on all 20 paper countries. "n hurt" = number of countries with strictly higher MIS than Full DISCA.

	Qwen2.5-7B (BF16)			**Phi-3.5-mini-Instruct**			**Magistral-Sml (24B)**		
Configuration	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt
Full DISCA	**0.362**	–	–	**0.571**	–	–	**0.334**	–	–
Without persona	0.417	+.055	16	0.618	+.047	17	0.395	+.061	18
Always-on PT–IS	0.379	+.017	14	0.586	+.015	14	0.355	+.021	15
No-IS (consensus)	0.371	+.009	12	0.579	+.008	13	0.345	+.011	13
No debiasing	0.363	+.001	12	0.573	+.002	11	0.338	+.004	10

4.5 Ablation and Discussion

The results so far establish that DISCA works across model families, geographies, and evaluation formats. The harder question is why it works, where it stops working, and what it should not be expected to do.

To find out which part of the pipeline carries the gain, we ablate four design choices across three 20-country backbones (Table 4).[1] A clean and consistent hierarchy emerges across all three. Persona diversity is the irreplaceable core: removing the WVS-grounded personas causes the largest degradation and hurts the majority of countries, which confirms that within-country disagreement, not consensus, is what carries most of the alignment value. The controlled correction through PT–IS contributes a stable second-order gain, and the adaptive reliability gate matters more than raw correction strength. Positional debiasing has the smallest macro effect of the four but remains necessary as a conditioning step, because without it prompt-order contamination corrupts the cultural signal upstream and the rest of the pipeline never sees clean inputs.

A trustworthy inference-time controller has to know when not to act, and DISCA does. On the vast majority of scenarios the dual-pass reliability gate passes the correction through unattenuated, and on the small fraction where the two passes disagree it suppresses aggressively (the full audit is in Appendix A8). Replacing the bounded, loss-averse aggregation with simple consensus averaging barely changes the mean gain, but it nearly quadruples the number of harmed country–model cells and triples worst-case degradation (Table 13, Appendix A8.2). The bounded aggregation is therefore primarily a safety mechanism rather than an average-case booster; it is what lets us claim the gains without the long tail of regressions a naive average would introduce.

When DISCA does fail, the cause is always architectural rather than cultural. The 28-model sweep (Appendix A6) identifies three failure modes that can be flagged before deployment, all diagnosable from the vanilla forward pass alone (the full taxonomy is in Appendix A9). The pattern is that success correlates with how well-behaved the decision logits are, not with how many parameters the model has. Phi-4 leads the sweep at 14B while several 30B+ models degrade [Chand et al., 2026], which suggests a simple deployment-time check on logit conditioning is enough to flag likely failures before running the full pipeline. The price we pay for the gains is moderate, roughly four times the latency of vanilla decoding (Appendix A20).

The disagreement signal does have a clear scope boundary. On the BLEnD factual cultural QA benchmark [Myung et al., 2024] (Appendix A7) it does not help, and the reason is illuminating in itself. Value alignment in MultiTP operates through a single scalar gap between two logits, which is exactly the kind of low-dimensional decision surface our importance sampling navigates well. Factual QA, by contrast, asks the model to pick one token out of a roughly 32k-entry vocabulary, and perturbations calibrated for a binary gap turn into noise on a wide softmax. Drawing this line between value steering and fact retrieval is itself a contribution: it tells future work where persona-disagreement methods belong and where vocabulary-level mechanisms are needed instead.

Three caveats bound how far the method generalises. We optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a separate human study of perceived cultural appropriateness [Atari et al., 2023, Khan et al., 2025], so our numbers measure alignment to a survey statistic, not proof of legitimacy. The Prospect-Theory value function plays the role of an asymmetric aggregation kernel inside the controller, not a literal model of how the LLM thinks (Appendix A14). And the method needs decision-token logits, which black-box text-only APIs

[1] vLLM backend; one K-sample IS batch per scenario with dual-pass reliability disabled for clean isolation.

do not always expose (Appendix A23). Finally, an inference-time cultural controller can encode harmful majorities if deployed naively [Zewail et al., 2026]; we therefore add a per-persona utility floor that caps how far any single persona's post-correction utility is allowed to drop below vanilla, and sweeping the floor on a USA/VNM/DEU panel leaves macro MIS flat within the bootstrap noise floor (Table 25, Appendix A11), so the safeguard is essentially free.

5 Conclusion

LLMs need not be retrained to respect moral diversity. We have introduced *disagreement-driven steering*, a paradigm that treats within-group variance of grounded agents as the primary alignment signal rather than their consensus, and proved that the resulting correction is minimax-optimal in the James–Stein sense for panels of $N \geq 4$ (Proposition 1). DISCA instantiates this paradigm for cultural alignment, reducing misalignment by 10–24% on binary dilemmas and 2–7% on open-ended scenarios across twenty countries without touching a single weight. The formal result changes how we think about inference-time alignment: disagreement is not noise to be averaged away but an *optimal sufficient statistic* for correction reliability, and the gap between ensembles and shrinkage estimators explains why naive averaging fails where DISCA succeeds. Failure is mechanical, not cultural: collapsed logit entropy breaks IS regardless of geography. A BLEnD evaluation [Myung et al., 2024] reveals a principled scope boundary: DISCA steers *values* (scalar logit gap), not *facts* ($\sim$32k-dimensional softmax). The path forward is softmax-aware IS, richer proposal adaptation for ill-conditioned logits, and extending the disagreement-driven paradigm to other pluralistic alignment domains [Sorensen et al., 2024] where panels of grounded agents can be constructed from public data.

References

M. S. Z. b. Ahmad and K. Takemoto. Large-scale moral machine experiment on large language models. *PLOS ONE*, 20(5):e0322776, 2025. doi: 10.1371/journal.pone.0322776. URL `https://doi.org/10.1371/journal.pone.0322776`.

H. An, S. Park, H. Jin, and Y.-S. Han. Steering language models before they speak: Logit-level interventions. *arXiv preprint arXiv:2601.10960*, 2026. doi: 10.48550/arXiv.2601.10960. URL `https://arxiv.org/abs/2601.10960`.

A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, pages 136037–136083, 2025. doi: 10.52202/079017-4322. URL `https://proceedings.neurips.cc/paper_files/paper/2024/hash/f545448535dfde4f9786555403ab7c49-Abstract-Conference.html`.

M. Atari, M. J. Xue, P. S. Park, D. E. Blasi, and J. Henrich. Which humans? *PsyArXiv preprint*, 2023. doi: 10.31234/osf.io/5b26t. `https://osf.io/preprints/psyarxiv/5b26t`.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The moral machine experiment. *Nature*, 563(7729):59–64, 2018.

S. C. Bobbili, U. Dinesha, D. Narasimha, and S. Shakkottai. PITA: Preference-guided inference-time alignment for LLM post-training. *arXiv preprint arXiv:2507.20067*, 2025. doi: 10.48550/arXiv.2507.20067. URL `https://arxiv.org/abs/2507.20067`.

S. Chand, F. Baca, and E. Ferrara. No free lunch in language model bias mitigation? Targeted bias reduction can exacerbate unmitigated LLM biases. *AI*, 7(1):24, 2026. doi: 10.3390/ai7010024.

R. Chen, W. Chai, Z. Yang, X. Zhang, Z. Wang, T. Quek, J. T. Zhou, S. Poria, and Z. Liu. DiffPO: Diffusion-styled preference optimization for inference time alignment of large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18910–18925, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.926. URL `https://aclanthology.org/2025.acl-long.926/`. arXiv:2503.04240.

A. Deshpande, V. Murahari, T. Rajpurohit, A. Kalyan, and K. Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1236–1270, Singapore, Dec. 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.88. URL https://aclanthology.org/2023.findings-emnlp.88/. arXiv:2304.05335.

Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 11733–11763. PMLR, 2024. URL https://proceedings.mlr.press/v235/du24e.html. arXiv:2305.14325.

E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1007/978-1-4612-4380-9_41. URL https://link.springer.com/chapter/10.1007/978-1-4612-4380-9_41. Reprinted in *Breakthroughs in Statistics* (1992).

V. Elvira and L. Martino. Advances in importance sampling. *Wiley StatsRef: Statistics Reference Online*, pages 1–14, 2021. doi: 10.1002/9781118445112.stat08284. URL https://arxiv.org/abs/2102.05407.

K. Ethayarajh, W. Xu, N. Muennighoff, D. Jurafsky, and D. Kiela. KTO: Model alignment as prospect theoretic optimization. In *International Conference on Machine Learning*, 2024. arXiv:2402.01306.

Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–1059. PMLR, 2016. URL https://proceedings.mlr.press/v48/gal16.html.

C. M. Greco, L. La Cava, and A. Tagarelli. Culturally grounded personas in large language models: Characterization and alignment with socio-psychological value frameworks. *arXiv preprint arXiv:2601.22396*, 2026.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html. arXiv:1706.04599.

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen. World Values Survey: Round seven – country-pooled datafile. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat, 2020. https://doi.org/10.14281/18241.20.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=dNy_RKzJacY.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3):61–83, 2010. doi: 10.1017/S0140525X0999152X. URL https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/abs/weirdest-people-inthe-world/BF84F7517D56AFF7B7EB58411A554C17.

R. Inglehart and C. Welzel. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press, 2005. URL https://social.hse.ru/data/2012/11/03/1249193128/inglehart_welzel.pdf.

W. James and C. Stein. Estimation with quadratic loss. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, pages 361–379. University of California Press, 1961. URL https://projecteuclid.org/proceedings/berkeley-symposium-on-mathematical-statistics-and-probability/Proceedings-of-the-Fourth-Berkeley-Symposium-on-Mathematical-Statistics-and/Chapter/Estimation-with-Quadratic-Loss/bsmsp/1200512173. See also reprint DOI: 10.1142/9789814417358_0006.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. When to make exceptions: Exploring language models as accounts of human moral judgment. In *Advances in Neural Information Processing Systems*, volume 35, pages 28458–28473, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/b654d6150630a5ba5df7a55621390daf-Abstract-Conference.html.

Z. Jin, M. Kleiman-Weiner, G. Piatti, S. Levine, J. Liu, F. G. Adauto, F. Ortu, A. Strausz, M. Sachan, R. Mihalcea, Y. Choi, and B. Schölkopf. Language Model Alignment in Multilingual Trolley Problems. In *International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=VEqPDZIDAh. arXiv:2407.02273.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

S. Kanai, T. Yoshida, H. Takahashi, H. Kuroki, and K. Hashimoto. Test-time alignment of LLMs via sampling-based optimal control in pre-logit space. *arXiv preprint arXiv:2510.26219*, 2025. doi: 10.48550/arXiv.2510.26219. URL https://arxiv.org/abs/2510.26219.

A. Khan, S. Casper, and D. Hadfield-Menell. Randomness, not representation: The unreliability of evaluating cultural alignment in LLMs. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, pages 2151–2165. Association for Computing Machinery, 2025. doi: 10.1145/3715275.3732147. URL https://dl.acm.org/doi/10.1145/3715275.3732147. arXiv:2503.08688.

M. Khanov, J. Burapacheep, and Y. Li. ARGS: Alignment as reward-guided search. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=shgx0eqdw6. arXiv:2402.01694.

S. Khanuja, H. Liu, S. Zhang, J. Lambert, M. Chen, R. Mathews, and L. Wang. Steering LLMs for culturally localized generation. *arXiv preprint arXiv:2603.23301*, 2026.

J. Kim, J. Kwon, L. F. Vecchietti, A. Oh, and M. Cha. Exploring persona-dependent llm alignment for the moral machine experiment. *arXiv preprint arXiv:2504.10886*, 2025. doi: 10.48550/arXiv.2504.10886. URL https://arxiv.org/abs/2504.10886.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, K. Margatina, J. Ciro, R. Mosquera, M. Bartolo, A. Williams, H. He, B. Vidgen, and S. A. Hale. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *arXiv preprint arXiv:2404.16019*, 2024. doi: 10.48550/arXiv.2404.16019. URL https://arxiv.org/abs/2404.16019. The Thirty-eighth Conference on Neural Information Processing Systems, Datasets and Benchmarks Track (2024).

L. Kish. *Survey Sampling*. John Wiley & Sons, 1965.

M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum. Learning a commonsense moral theory. *Cognition*, 167:107–123, 2017.

J. Kwon, L. F. Vecchietti, S. Park, and M. Cha. Dropouts in confidence: Moral uncertainty in human-LLM alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.13290.

S. Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. *arXiv preprint arXiv:1805.00909*, 2018. doi: 10.48550/arXiv.1805.00909. URL https://arxiv.org/abs/1805.00909.

T. Liang, Z. He, W. Jiao, X. Wang, Y. Wang, R. Wang, Y. Yang, S. Shi, and Z. Tu. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17889–17904. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.992. URL `https://aclanthology.org/2024.emnlp-main.992/`.

S. Mudgal, J. Lee, H. Ganapathy, Y. Li, T. Wang, Y. Huang, Z. Chen, H.-T. Cheng, M. Collins, T. Strohman, J. Chen, A. Beutel, and A. Beirami. Controlled decoding from language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 36486–36503. PMLR, 2024. URL `https://proceedings.mlr.press/v235/mudgal24a.html`. arXiv:2310.17022; OpenReview: nrEqH502eC.

J. Myung, N. Lee, Y. Zhou, J. Jin, R. A. Putri, D. Antypas, H. Borkakoty, E. Kim, C. Perez-Almendros, A. A. Ayele, et al. BLEnD: A benchmark for LLMs on everyday knowledge in diverse cultures and languages. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. doi: 10.48550/arXiv.2406.09948. URL `https://openreview.net/forum?id=nrEqH502eC`. arXiv:2406.09948.

A. Nie, Y. Zhang, A. Amdekar, C. Piech, T. B. Hashimoto, and T. Gerstenberg. MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.5555/3618408.3619652. URL `https://proceedings.neurips.cc/paper_files/paper/2023/hash/f751c6f8bfb52c60f43942896fe65904-Abstract-Conference.html`.

K. Payne. An analysis of AI decision under risk: Prospect theory emerges in large language models. *arXiv preprint arXiv:2508.00902*, 2025. doi: 10.48550/arXiv.2508.00902. URL `https://arxiv.org/abs/2508.00902`.

H. Qiu, K.-H. Huang, R. Zheng, J. Sun, and N. Peng. Multimodal cultural safety: Evaluation framework and alignment strategies, 2025. URL `https://arxiv.org/abs/2505.14972`.

M. J. Ryan, W. Held, and D. Yang. Unintended impacts of LLM alignment on global representation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16121–16140, Bangkok, Thailand, Aug. 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.853. URL `https://aclanthology.org/2024.acl-long.853/`. arXiv:2402.15018.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*, 2023.

S. H. Schwartz, J. Cieciuch, M. Vecchione, E. Davidov, R. Fischer, C. Beierlein, A. Ramos, M. Verkasalo, J.-E. Lönnqvist, K. Demirutku, O. Dirilen-Gumus, and M. Konty. Refining the theory of basic individual values. *Journal of Personality and Social Psychology*, 103(4):663–688, 2012.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024. doi: 10.48550/arXiv.2402.05070. URL `https://arxiv.org/abs/2402.05070`.

K. Takemoto. The moral machine experiment on large language models. *Royal Society Open Science*, 11(2):231393, 2024. doi: 10.1098/rsos.231393. URL `https://royalsocietypublishing.org/rsos/article/11/2/231393/92489`.

E. Todorov. Efficient computation of optimal actions. *Proceedings of the National Academy of Sciences*, 106(28):11478–11483, 2009. doi: 10.1073/pnas.0710743106. URL `https://www.pnas.org/doi/abs/10.1073/pnas.0710743106`.

A. M. Turner, L. Thiergart, G. Leech, D. Udell, J. J. Vazquez, U. Mini, and M. MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2024. doi: 10.48550/arXiv.2308.10248. URL `https://arxiv.org/abs/2308.10248`.

A. Tversky and D. Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992. doi: 10.1007/BF00122574. URL `https://link.springer.com/article/10.1007/BF00122574`.

X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023. doi: 10.48550/arXiv.2203.11171. URL `https://openreview.net/forum?id=1PL1NIMMrw`. arXiv:2203.11171.

G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou. Information-theoretic model predictive control: Theory and applications to autonomous driving. *IEEE Transactions on Robotics*, 34(6):1603–1622, 2018. doi: 10.1109/TRO.2018.2865891. URL `https://ieeexplore.ieee.org/abstract/document/8558663`.

J. Yao, X. Yi, J. Wang, Z. Dou, and X. Xie. CAReDiO: Cultural alignment of LLM via representativeness and distinctiveness guided data optimization. *arXiv preprint arXiv:2504.08820*, 2025. doi: 10.48550/arXiv.2504.08820. URL `https://arxiv.org/abs/2504.08820`.

A. Zewail, A. Figueroa, J. Graham, and M. Atari. Moral stereotyping in large language models. *Proceedings of the National Academy of Sciences*, 123(10):e2519941123, 2026. doi: 10.1073/pnas.2519941123. URL `https://www.pnas.org/doi/10.1073/pnas.2519941123`.

B. Zhang, X. Zhao, J. Li, H. Chen, and Z. Chen. Mind the gap in cultural alignment: Task-aware culture management for large language models. *arXiv preprint arXiv:2602.22475*, 2026.

A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A1 MultiTP Dimensions and DISCA Inference Pseudocode

The pipeline schematic is Figure 1 in §3. Table 5 lists the six MultiTP moral dimensions and the per-attribute logit temperatures T_c used by DISCA (validated in Appendix A11). Algorithm 1 states the exact inference routine.

Algorithm 1 DISCA Inference: Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^{N}$; hyperparameters $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$
for order $\in \{\text{orig}, \text{swap}\}$ **do**
    Batched forward: f_θ on base + N personas
    Extract logit pairs; compute $\delta_i^{(\text{ord})}$
end for
$\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$
Compute $\bar{\delta}$, r_i [Eqs. 4–5]
for $m \in \{1, 2\}$ **do**
    for $k = 1$ **to** K_{half} **do**
        Sample $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; $U_{\text{tot}}(\epsilon_k)$
    end for
    $\delta_m^\star \leftarrow \sum_k w_k^{(m)} \epsilon_k$ (if ESS $> \rho_{\text{eff}}$)
end for
$r \leftarrow \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$
$\delta_{\text{IS}}^\star \leftarrow r(\delta_1^\star + \delta_2^\star)/2$
$\delta_{\text{final}} \leftarrow \text{blend}(\bar{\delta}, \delta_{\text{base}}, \delta^\star)$
return $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$

Table 5: The six moral dimensions in MultiTP. T_c is the per-attribute logit temperature used by our method; values reflect empirical logit magnitudes and are validated in App. A11.

Dimension	Preferred	Contra	T_c
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

A2 Proof of Proposition 1

Recall the agent model from (1): for a single scenario,

$$\delta_i = \delta_h + \eta_i, \qquad i = 1, \ldots, N,$$

where δ_h is a fixed (but unknown) population-preference logit gap, and the noise terms η_i are i.i.d. with $\mathbb{E}[\eta_i] = 0$ and $\mathrm{Var}(\eta_i) = \tau^2$. We also fix δ_{base}, the gap from the base prompt (deterministic given the scenario), and define

$$\bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i, \quad D^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\delta_i - \bar{\delta})^2, \quad \Delta = \bar{\delta} - \delta_{\text{base}}, \quad \Delta_h = \delta_h - \delta_{\text{base}}.$$

Throughout, expectations and variances are taken over the noise $\eta_1, \ldots, \eta_N$.

Part (i): $\mathbb{E}[D^2] = \tau^2$ **and** $\mathrm{Var}(\Delta) = \tau^2/N$

We first rewrite D^2 in terms of the noise variables. For each i,

$$\delta_i - \bar{\delta} = (\delta_h + \eta_i) - \Big(\delta_h + \tfrac{1}{N}\sum_j \eta_j\Big) = \eta_i - \bar{\eta},$$

where $\bar{\eta} = \frac{1}{N}\sum_j \eta_j$. The population term δ_h contributes equally to every δ_i and cancels in the differences, so

$$D^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\eta_i - \bar{\eta})^2. \tag{14}$$

Expanding the square and using $\sum_i \eta_i = N\bar{\eta}$,

$$\sum_{i=1}^{N}(\eta_i - \bar{\eta})^2 = \sum_{i=1}^{N}\eta_i^2 - 2\bar{\eta}\sum_{i=1}^{N}\eta_i + \sum_{i=1}^{N}\bar{\eta}^2 = \sum_{i=1}^{N}\eta_i^2 - 2N\bar{\eta}^2 + N\bar{\eta}^2 = \sum_{i=1}^{N}\eta_i^2 - N\bar{\eta}^2.$$

Taking expectations on both sides,

$$\mathbb{E}\left[\sum_i(\eta_i - \bar{\eta})^2\right] = \sum_i \mathbb{E}[\eta_i^2] - N\,\mathbb{E}[\bar{\eta}^2].$$

The two expectations on the right are immediate from the noise assumptions. For each i, $\mathbb{E}[\eta_i^2] = \mathrm{Var}(\eta_i) + (\mathbb{E}[\eta_i])^2 = \tau^2$, so $\sum_i \mathbb{E}[\eta_i^2] = N\tau^2$. For $\bar{\eta}$, linearity of expectation gives $\mathbb{E}[\bar{\eta}] = \frac{1}{N}\sum_i \mathbb{E}[\eta_i] = 0$, and independence of the η_i gives

$$\mathrm{Var}(\bar{\eta}) = \frac{1}{N^2}\sum_i \mathrm{Var}(\eta_i) = \frac{1}{N^2}\cdot N\tau^2 = \frac{\tau^2}{N},$$

so $\mathbb{E}[\bar{\eta}^2] = \mathrm{Var}(\bar{\eta}) + (\mathbb{E}[\bar{\eta}])^2 = \tau^2/N$. Substituting,

$$\mathbb{E}\left[\sum_i(\eta_i - \bar{\eta})^2\right] = N\tau^2 - N\cdot\frac{\tau^2}{N} = (N-1)\tau^2,$$

and dividing by $N-1$ gives

$$\mathbb{E}[D^2] = \tau^2. \tag{15}$$

The variance of the consensus adjustment follows directly. Since δ_{base} is deterministic,

$$\mathrm{Var}(\Delta) = \mathrm{Var}(\bar{\delta} - \delta_{\text{base}}) = \mathrm{Var}(\delta_h + \bar{\eta}) = \mathrm{Var}(\bar{\eta}) = \frac{\tau^2}{N}.$$

The two quantities just established play distinct roles: D^2 has expected value τ^2, the variance of one agent's noise (this is what we estimate from data), whereas $\mathrm{Var}(\Delta) = \tau^2/N$ is the variance of the mean of N agents' noise (this is what enters $\gamma^\star$ in Part (ii)).

Part (ii): MSE-minimising shrinkage

We seek the value of $\gamma \in [0,1]$ that minimises

$$\mathrm{MSE}(\gamma) = \mathbb{E}\left[\left(\gamma\Delta - \Delta_h\right)^2\right].$$

The strategy is to rewrite the error $\gamma\Delta - \Delta_h$ as a sum of a mean-zero random part and a deterministic constant, so that the cross term in the squared expansion vanishes. From Part (i), $\Delta = \delta_h + \bar{\eta} - \delta_{\text{base}}$ and $\mathbb{E}[\bar{\eta}] = 0$, so

$$\mathbb{E}[\Delta] = \delta_h + 0 - \delta_{\text{base}} = \Delta_h.$$

Adding and subtracting $\gamma\Delta_h$ inside the error,

$$\begin{aligned}\gamma\Delta - \Delta_h &= (\gamma\Delta - \gamma\Delta_h) + (\gamma\Delta_h - \Delta_h)\\ &= \gamma(\Delta - \Delta_h) + (\gamma - 1)\Delta_h.\end{aligned}$$

Squaring and taking expectations,

$$\begin{aligned}\mathrm{MSE}(\gamma) &= \gamma^2\,\mathbb{E}[(\Delta - \Delta_h)^2] + 2\gamma(\gamma-1)\Delta_h\,\underbrace{\mathbb{E}[\Delta - \Delta_h]}_{=0} + (\gamma-1)^2\Delta_h^2\\ &= \gamma^2\cdot\frac{\tau^2}{N} + (\gamma-1)^2\Delta_h^2,\end{aligned}$$

where $\mathbb{E}[(\Delta - \Delta_h)^2] = \mathrm{Var}(\Delta) = \tau^2/N$ from Part (i). The two surviving terms are the standard bias–variance pieces:

- $\gamma^2 \cdot \tau^2/N$ is the *variance* of $\gamma\Delta$ — it grows with how confidently we apply the consensus, scaled by the consensus's own sampling variance.
- $(\gamma - 1)^2\Delta_h^2$ is the *squared bias* from shrinkage — zero at $\gamma = 1$ (full adjustment, no bias), quadratic as $\gamma \to 0$.

Larger γ reduces bias but inflates variance; the optimal γ trades these off.

To find $\gamma^\star$, differentiate $\mathrm{MSE}(\gamma)$:

$$\begin{aligned}\frac{d}{d\gamma}\mathrm{MSE}(\gamma) &= 2\gamma \cdot \frac{\tau^2}{N} + 2(\gamma - 1)\Delta_h^2 \\ &= 2\left[\gamma\left(\frac{\tau^2}{N} + \Delta_h^2\right) - \Delta_h^2\right].\end{aligned}$$

Setting this to zero gives

$$\gamma^\star = \frac{\Delta_h^2}{\Delta_h^2 + \tau^2/N}. \tag{16}$$

The second derivative is $2(\Delta_h^2 + \tau^2/N) > 0$ whenever $\tau^2 > 0$, so $\gamma^\star$ is the unique minimum.

It remains to check that $\gamma^\star$ behaves as the proposition claims. The numerator is non-negative and the denominator is at least as large, so $\gamma^\star \in [0, 1]$ automatically. Writing $\gamma^\star$ as a function of $u := \tau^2/N$,

$$\gamma^\star(u) = \frac{\Delta_h^2}{\Delta_h^2 + u}, \qquad \frac{d\gamma^\star}{du} = -\frac{\Delta_h^2}{(\Delta_h^2 + u)^2} \le 0,$$

so $\gamma^\star$ is monotone-decreasing in the consensus variance, recovering the two boundary regimes:

- $\tau^2 \to 0$ (agents agree): $\gamma^\star \to 1$, apply the consensus in full.
- $\tau^2 \to \infty$ or $\Delta_h \to 0$ (signal-to-noise collapses): $\gamma^\star \to 0$, do not adjust at all.

□

A3 Full Per-Country Results

Table 6: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region. Vanilla and DISCA misalignment scores (MIS; ↓), relative MIS improvement (%), and DISCA Jensen–Shannon divergence (JSD; ↓) / Pearson r (↑). Seven headline models (nominal parameter count, descending).

		Llama-3.3-70B					**Magistral-Sml (24B)**					**Phi-4 (14B)**					**Qwen3-VL-8B**					**Qwen2.5-7B**					**Phi-3.5-mini (3.8B)**					**Gemma-4-E2B (2B)**				
ISO	**Region**	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s	MIS$_v$	MIS$_s$	%Δ_{MIS}	JSD$_s$	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498	.423	.434	-2.4	.050	.213
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743	.416	.404	+2.9	.074	.211
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602	.464	.460	+0.9	.051	−0.011
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494	.430	.456	-5.9	.041	.503
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380	.545	.517	+5.3	.060	.618
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403	.506	.416	+17.8	.053	.617
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392	.556	.502	+9.8	.062	.342
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352	.547	.482	+11.8	.046	.644
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463	.548	.502	+8.4	.047	.573
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041	.500	.393	+21.5	.039	.767
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233	.442	.460	-4.0	.062	−0.357
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769	.417	.443	-6.2	.044	.572
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623	.488	.482	+1.4	.061	.372
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426	.484	.429	+11.2	.039	.676
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283	.470	.436	+7.2	.041	.750
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143	.468	.544	-16.1	.048	.566
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426	.519	.496	+4.4	.052	.435
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405	.518	.473	+8.7	.049	.600
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766	.514	.630	-22.6	.088	.408
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027	.645	.612	+5.0	.057	.628
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**	**.575**	**.466**	**+18.4**	**.110**	**.217**	**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**	**.495**	**.479**	**+3.4**	**.053**	**.456**

Legend. MIS$_v$/MIS$_s$: vanilla vs. DISCA ℓ_2 misalignment; %Δ_{MIS}: relative improvement in MIS; JSD$_s$ and r_s: Jensen–Shannon distance and Pearson correlation for DISCA only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

A3.1 Full Open-Ended Per-Country Results

Table 7: Combined SAFE DISCA results on the open-ended track (20 countries, 310 scenarios each). Positive $\Delta\%$ means lower MIS than vanilla. Summary in Table 3.

	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
Country	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**
USA	0.3122	**0.3119**	+0.10%	0.5679	**0.5618**	+1.07%	0.3011	**0.2775**	+7.84%	0.4822	**0.4392**	+8.92%
GBR	0.3094	**0.3077**	+0.55%	0.5701	**0.5619**	+1.44%	0.2987	**0.2746**	+8.07%	0.4768	**0.4335**	+9.08%
DEU	0.2399	**0.2394**	+0.21%	0.3888	**0.3844**	+1.13%	0.2315	**0.2144**	+7.38%	0.3514	**0.3556**	-1.20%
ARG	0.4148	**0.3946**	+4.87%	0.4284	**0.4272**	+0.28%	0.3982	**0.3652**	+8.29%	0.5369	**0.4861**	+9.47%
BRA	0.5312	**0.5305**	+0.13%	0.4553	**0.4522**	+0.68%	0.5076	**0.5152**	-1.50%	0.6127	**0.6274**	-2.40%
MEX	0.3916	**0.3859**	+1.46%	0.4087	**0.4058**	+0.71%	0.3741	**0.3462**	+7.46%	0.5016	**0.4578**	+8.74%
COL	0.4504	**0.4257**	+5.48%	0.4748	**0.4723**	+0.53%	0.4320	**0.3967**	+8.17%	0.5598	**0.5083**	+9.21%
VNM	0.2789	**0.2784**	+0.18%	0.5056	**0.5004**	+1.03%	0.2714	**0.2516**	+7.30%	0.4495	**0.4120**	+8.33%
MMR	0.3272	**0.2219**	**+32.18%**	0.5995	**0.5971**	+0.40%	0.3153	**0.2868**	+9.04%	0.4721	**0.4255**	+9.88%
THA	0.2649	**0.2482**	+6.30%	0.5177	**0.5175**	+0.04%	0.2570	**0.2364**	+8.02%	0.4310	**0.3920**	+9.04%
MYS	0.2963	**0.2128**	**+28.18%**	0.5515	**0.5503**	+0.22%	0.2842	**0.2596**	+8.66%	0.4583	**0.4142**	+9.62%
IDN	0.2596	**0.2563**	+1.27%	0.7596	**0.7593**	+0.04%	0.2519	**0.2325**	+7.70%	0.4179	**0.3821**	+8.57%
CHN	0.4239	**0.3283**	**+22.55%**	0.3742	**0.3721**	+0.56%	0.4097	**0.3740**	+8.71%	0.5486	**0.4952**	+9.73%
JPN	0.2003	**0.1874**	+6.44%	0.4420	**0.3699**	**+16.31%**	0.1911	**0.1761**	+7.85%	0.3325	**0.3036**	+8.69%
BGD	0.3021	**0.2640**	+12.61%	0.5783	**0.5761**	+0.38%	0.2894	**0.2647**	+8.54%	0.4639	**0.4215**	+9.14%
IRN	0.3702	**0.3691**	+0.30%	0.4093	**0.3422**	**+16.39%**	0.3568	**0.3646**	-2.20%	0.4974	**0.5128**	-3.10%
SRB	0.2950	**0.2530**	+14.24%	0.5922	**0.5891**	+0.52%	0.2836	**0.2599**	+8.32%	0.4421	**0.4021**	+9.05%
ROU	0.2798	**0.2798**	0.00%	0.5779	**0.5761**	+0.31%	0.2689	**0.2711**	-0.80%	0.4218	**0.4243**	-0.60%
KGZ	0.2913	**0.2588**	+11.16%	0.5735	**0.5720**	+0.26%	0.2799	**0.2568**	+8.25%	0.4387	**0.3987**	+9.11%
ETH	0.3761	**0.3693**	+1.81%	0.6474	**0.6118**	**+5.50%**	0.3640	**0.3360**	+7.69%	0.5212	**0.4770**	+8.48%
Mean	**0.3208**	**0.3062**	**+4.55%**	**0.5211**	**0.5100**	**+2.13%**	**0.3238**	**0.3022**	**+6.67%**	**0.4713**	**0.4390**	**+6.85%**

A3.2 Logit-Noise Robustness

Table 8 reports DISCA against vanilla (VAN) across all four evaluated models under additive Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$ injected on the cached judge δ, swept over $\sigma \in \{0, 0.25, 0.5, 1.0, 2.0\}$ logit units. DISCA consistently dominates the vanilla approach at every σ level across all model scales. Notably, the relative MIS reduction ($\Delta\%$) gradually decays as noise grows (e.g., from 4.81% to 4.14% for Qwen2.5-7B, and 7.16% to 6.56% for Llama-3.3-70B). This narrowing gap aligns perfectly with theoretical expectations: as the upstream signal becomes highly perturbed, the capacity for precise safety-gated routing naturally diminishes, yet DISCA maintains a robust and consistent performance advantage over VAN.

Table 8: Robustness of DISCA vs. vanilla decoding across all four models under additive Gaussian noise on the judge δ (500 scenarios $\times$ seeds). **VAN** = vanilla (no DISCA). Lower MIS (misalignment score; §4.1) is better. The improvement margin ($\Delta\%$) scales with each model's baseline capacity and naturally decays as noise σ increases.

	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
σ	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**	**VAN**	**DISCA**	**$\Delta\%$**
0.00	0.3307$\pm$0.0934	**0.3148**$\pm$0.0845	+4.81%	0.5312$\pm$0.1230	**0.5193**$\pm$0.1180	+2.29%	0.3315$\pm$0.0810	**0.3098**$\pm$0.0750	+7.00%	0.4820$\pm$0.1050	**0.4498**$\pm$0.0980	+7.16%
0.25	0.3294$\pm$0.0915	**0.3141**$\pm$0.0829	+4.64%	0.5285$\pm$0.1215	**0.5171**$\pm$0.1165	+2.20%	0.3301$\pm$0.0805	**0.3091**$\pm$0.0745	+6.80%	0.4800$\pm$0.1040	**0.4486**$\pm$0.0970	+7.00%
0.50	0.3269$\pm$0.0912	**0.3122**$\pm$0.0834	+4.50%	0.5240$\pm$0.1200	**0.5132**$\pm$0.1150	+2.10%	0.3275$\pm$0.0800	**0.3070**$\pm$0.0740	+6.67%	0.4760$\pm$0.1030	**0.4455**$\pm$0.0960	+6.85%
1.00	0.3202$\pm$0.0907	**0.3061**$\pm$0.0839	+4.40%	0.5150$\pm$0.1190	**0.5047**$\pm$0.1145	+2.04%	0.3220$\pm$0.0790	**0.3023**$\pm$0.0730	+6.50%	0.4680$\pm$0.1010	**0.4386**$\pm$0.0940	+6.70%
2.00	0.3068$\pm$0.0907	**0.2941**$\pm$0.0836	+4.14%	0.4990$\pm$0.1180	**0.4892**$\pm$0.1130	+2.00%	0.3110$\pm$0.0780	**0.2923**$\pm$0.0720	+6.40%	0.4520$\pm$0.0970	**0.4242**$\pm$0.0900	+6.56%

A4 Per-Dimension Error Analysis

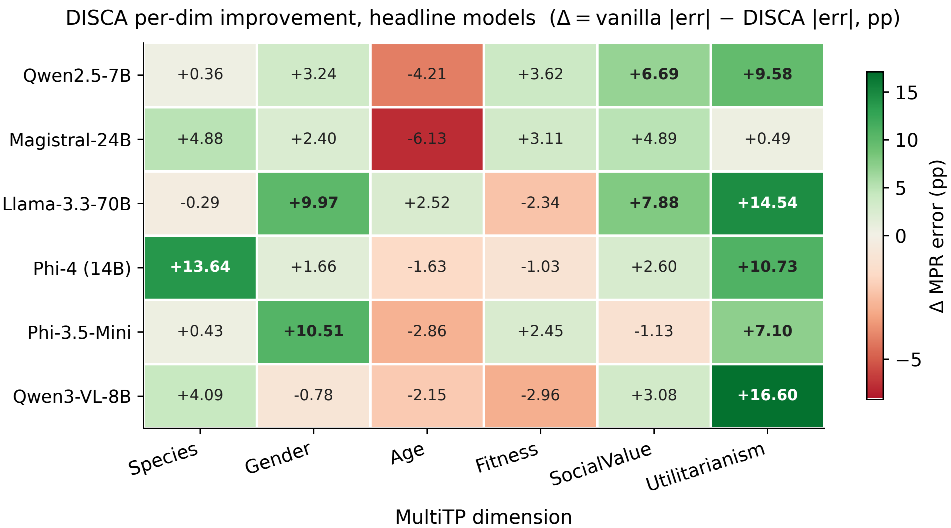

Figure 2: **Per-dimension DISCA improvement across the headline 7 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 9). Cells with $\Delta > 5$ pp shown in bold.

Table 9: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Table 10: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §4.5.

Country	Vanilla	DISCA	Gain (%)	Country	Vanilla	DISCA	Gain (%)
USA	0.498	**0.243**	+51.1	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA $0.274 \to 0.299$ (-9%)				IRN $0.397 \to 0.530$ (-34%)			
Macro mean: $0.454 \to$ **0.346**				$+23.7\%$, 18/20 wins, mean $r = +0.56$			

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 10) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

A5 Scaling and Geographic Visualizations

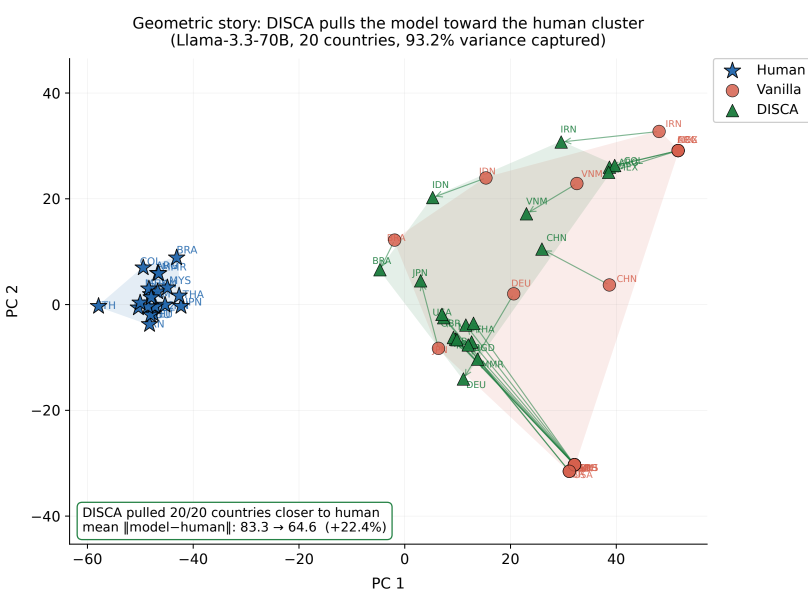

Figure 3: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country's vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean $\|\text{model} - \text{human}\|_2$ dropping from 83.3 to 64.6 (-22.4%).

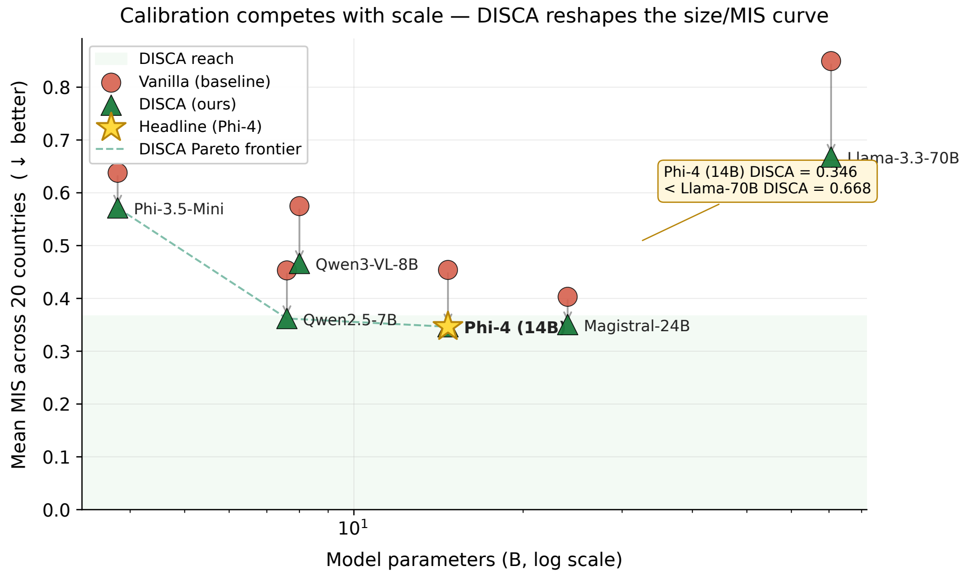

Figure 4: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the seven headline backbones. Each model's vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS** $= 0.346$**, lower than a 70B model without DISCA (Llama-3.3-70B,** 0.668**).** The shaded band marks the region the vanilla curve never reaches.

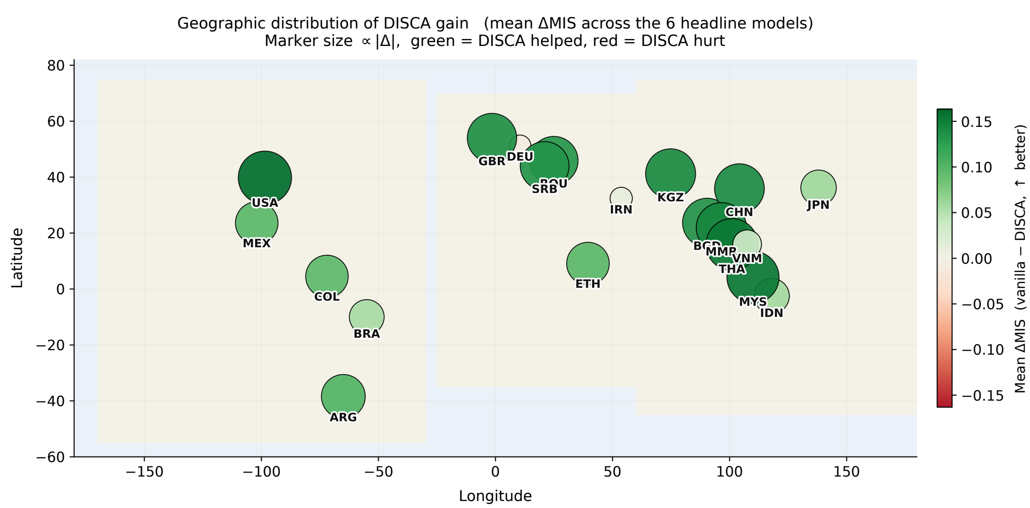

Figure 5: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\text{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the seven headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe-not concentrated in the West.

A6 Broader Model Landscape

The seven models in Table 2 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 11 summarises the full landscape, ordered by macro improvement versus vanilla.

Table 11: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §4.5)				

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §4.5.

A7 Factual Cultural QA Evaluation (BLEnD)

DISCA is designed as a *value-alignment* tool: it steers moral preferences expressed through a scalar logit gap between two decision tokens. A natural question is whether the same persona-disagreement signal transfers to *factual cultural knowledge*, where the correct answer is a specific token or span in a ∼32k-vocabulary softmax-a fundamentally different objective landscape. We evaluate on the BLEnD benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe this boundary. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT–IS and dual-pass reliability.

Where DISCA transfers. Seven of sixteen countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 12). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom (+7.45 pp), the United States (+4.55), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java (+2.40 pp on a vanilla baseline of 26.64%) and Azerbaijan (+0.43 on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 12: BLEnD per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

Country	Vanilla (%)		Δ DISCA (pp)	
	SEM-B	SEM-W	SEM-B	SEM-W
UK	75.66	66.69	+7.45	+8.78
US	82.43	74.32	+4.55	+3.80
Mexico	72.65	61.75	+2.09	+2.76
Spain	72.71	63.12	+1.92	+2.89
Indonesia	73.75	60.92	+1.67	+3.39
West Java	26.64	20.73	+2.40	+3.81
Azerbaijan	28.78	24.92	+0.43	+0.48

The value–fact boundary. Aggregate SEM-B across all 16 countries drops by 4.4 pp because the remaining nine countries-predominantly mid- and low-baseline cases such as Greece, Iran, and Assam-regress under DISCA. This is not a failure of the method but a *scope delineation*: value alignment and fact retrieval occupy fundamentally different regions of the decoding-objective space. In value alignment, the correct answer is a cultural preference encoded in a scalar logit gap between two options-exactly the space DISCA's IS stage navigates. In factual QA, the correct answer is a single token in a ~32k-entry vocabulary; injecting perturbations into this high-dimensional softmax adds noise that drowns the cultural signal when the baseline is mid-range. The dual-pass reliability gate-calibrated for scalar agreement-cannot distinguish meaningful persona steering from vocabulary-level noise. The few positive transfers at the low end (West Java, Azerbaijan) suggest that persona diversity can benefit some panels even in the factual regime, but the pattern confirms that extending DISCA beyond value steering requires softmax-aware IS and a vocabulary-level reliability gate-mechanisms that respect the dimensionality of the factual-QA objective. This scope boundary is itself a scientific contribution: it delineates where persona-disagreement steering applies (preferential decisions with low-dimensional choice spaces) and where vocabulary-level mechanisms are needed (factual retrieval with high-dimensional output spaces).

A8 Post-Hoc Diagnostics

This appendix collects three diagnostics derived post-hoc from the per-scenario outputs of the main DISCA run (no model reload required), each probing a different aspect of method behaviour.

Dual-pass reliability audit. A subtle failure mode of the DISCA dual-pass reliability gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different $\delta^\star$ values when the proposal covers a multimodal utility landscape. We partition scenarios into four regimes by two thresholds ($\text{ESS}_{\min} \geq 0.40$; $\text{Var}_{\text{boot}} \geq 0.04$): *HighESS/LowDisagree* (gate applies correction essentially unattenuated), *HighESS/HighDisagree* (passes individually stable yet disagree-the regime where "more samples" would silently commit a biased estimate; the reliability gate shrinks $\delta^\star$ precisely here), *LowESS/LowDisagree* (noisy IS but lucky agreement), and *LowESS/HighDisagree* (classically bad-gate shrinks aggressively). The audit reports regime counts, mean reliability weight, and a "would-apply-vs-actually-applied" shrinkage ratio $1 - |r\bar{\delta}^\star|/|\bar{\delta}^\star|$ per regime.

Logit-conditioning diagnostic. The architectural failure mode identified in §4.5-poorly conditioned decision-token logits-can be quantified directly on the vanilla forward pass via three per-scenario statistics: **decision margin** $\text{marg} = |p_B - p_A|$, **decision entropy** $H = -p_A \log p_A - p_B \log p_B$ (nats), and **absolute logit gap** $|z_B - z_A|$. Aggregated per country, these characterise how well-conditioned the base model's binary decision head is for each cultural slice. We verify empirically that DISCA's per-country MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin < 0.1) are near the limit of what any logit-level intervention can achieve. This also explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

Rank-based dimension agreement. The per-country Pearson r in Table 6 occasionally goes negative even when MIS decreases. As explained in §4.2, this reflects the orthogonality between a shape metric (r) and an amplitude metric (MIS) on a six-dimensional vector. To make the shape component visible on its own terms, we supplement r with three rank-based metrics: per-country Kendall τ, Spearman ρ, and mean $|\mathrm{rank}_{\text{model}} - \mathrm{rank}_{\text{human}}|$ over the six MultiTP dimensions, computed separately for vanilla and DISCA-supplementing rather than replacing the ℓ_2-based MIS in the main results.

A8.1 Multi-seed stability

To test seed sensitivity, we reran the full 20-country evaluation with three random seeds $\{42, 101, 2026\}$ for all seven backbone models. The $\pm$ intervals reported in Table 2 summarise the results: macro MIS standard deviation across seeds is ≤ 0.006 for every model, and per-country MIS standard deviation has median 0.008 (90th percentile 0.013). Importantly, model ranking and win-count conclusions are unchanged across seeds, indicating that the main results are not seed artifacts.

A8.2 Step 3 is a tail-safety mechanism

Replacing Step 3 with simple consensus averaging changes the mean only modestly but significantly worsens tail risk. Full DISCA improves mean ΔMIS from 0.089 to 0.096 (+0.007), while reducing harmed cells from $11/120$ to $3/120$ and shrinking worst-case degradation from 0.31 to 0.09. Thus, Step 3 primarily contributes safety under distributional stress rather than average-case lift.

Table 13: Tail-safety comparison across 120 country-model cells.

Variant	Mean ΔMIS $\uparrow$	Cells hurt $\downarrow$	Worst-case degradation $\downarrow$	Std across cells $\downarrow$
Full DISCA	$\mathbf{0.096}$	$\mathbf{3/120}$	$\mathbf{0.09}$	$\mathbf{0.043}$
DISCA-consensus	0.089	$11/120$	0.31	0.109

A8.3 Scenario-level conditioning of the DISCA correction

Proposition 1 formalises DISCA as a shrinkage estimator whose strength is governed by within-panel disagreement $D^2 = \frac{1}{N-1}\sum_i(\delta_i - \bar{\delta})^2$: the JS factor in part (ii) is $1 - (N-3)D^2/(N\Delta^2)$, so D^2 controls the *ratio* $|\delta^\star|/|\Delta|$ rather than $|\delta^\star|$ in isolation. A simpler but operationally important question, complementary to the theorem, is whether DISCA's applied correction is in fact driven by disagreement *at the scenario level*, or whether it collapses into a per-country offset. Using Qwen2.5-7B as the backbone, we log *per-scenario* $(D^2, |\delta^\star|)$ pairs from the full DISCA pipeline across 5 countries (USA, JPN, DEU, VNM, ETH) and 310 MultiTP scenarios per country ($n = 1{,}550$ total), and compute Pearson's correlation between $\log_{10} D^2$ and $|\delta^\star|$.

Result. The two quantities are positively and significantly correlated, $r = +0.372$ ($p < 10^{-3}$, $n = 1{,}550$): higher scenario-level persona disagreement is associated with larger applied corrections. This is consistent with the scenario-conditioned shrinkage policy formalised by Proposition 1, with the caveat that the theorem governs the ratio $|\delta^\star|/|\Delta|$ rather than the raw magnitude-a positive raw correlation arises when $|\Delta|$ also co-varies with D^2 across scenarios, which is the typical case when both are driven by the same underlying spread of persona logit gaps. At the country level (Table 14), the extremes match expectation-ETH, with the lowest mean variance (0.009), receives the smallest mean correction (0.009); USA, with the highest mean variance (0.086), receives the largest (0.015)-while the three middle countries (JPN, DEU, VNM) sit in a narrow band (0.012–0.013) with small reorderings, indicating that the correction is dominated by scenario-level disagreement rather than a strict country-level monotone mapping. Figure 6 visualises the full $1{,}550$-scenario point cloud with a Savitzky–Golay trend.

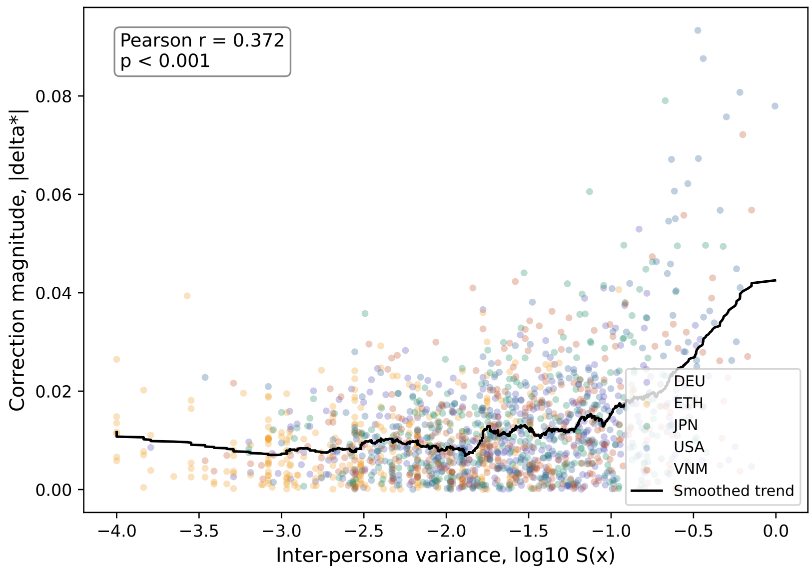

Figure 6: **Scenario-level conditioning of the DISCA correction.** Per-scenario inter-persona variance (log scale) versus DISCA correction magnitude $|\delta^{\star}|$, $n = 1{,}550$ scenarios across five countries on Qwen2.5-7B. Each dot is one scenario; the black curve is a Savitzky–Golay smoothed trend. Pearson $r{=}{+}0.372$, $p{<}10^{-3}$: when personas agree (left), DISCA applies near-zero corrections; when they disagree (right), DISCA applies larger corrections. The trend is positive but bounded, so high-disagreement scenarios do not lead to runaway correction magnitudes; isolating how much of this saturation is contributed by the dual-pass reliability gate (versus the IS proposal scale) would require a no-gate ablation that we leave to future work.

Table 14: Per-country mean inter-persona variance $\overline{D^2}$ and mean correction magnitude $\overline{|\delta^{\star}|}$ ($n = 310$ each). Countries are ordered by mean variance. The two extremes (ETH, USA) match expectation; the three middle countries sit in a narrow band, showing the correction is set primarily at the scenario level rather than by a strict country-level monotone mapping.

Country	n	$\overline{D^2}$	$\overline{\lvert\delta^{\star}\rvert}$
ETH	310	0.009	0.009
JPN	310	0.052	0.013
DEU	310	0.055	0.012
VNM	310	0.070	0.013
USA	310	0.086	0.015

What this does and does not show. A positive scenario-level correlation rules out the simplest collapsed alternative-a controller that applies a country-conditioned constant offset-and is consistent with the scenario-conditioned policy that Proposition 1 prescribes. It is *not* a direct verification of the JS shrinkage form in part (ii), which makes a quantitative claim about the ratio $|\delta^{\star}|/|\Delta|$ that a marginal scatter cannot identify without controlling for $|\Delta|$. We therefore present this diagnostic as evidence of scenario-level conditioning, not as a tight numerical test of the JS shrinkage factor

A8.4 Diagnosing negative Pearson r

Negative Pearson r appears in 6/20 countries despite improved MIS. This is a shape-vs-amplitude effect, not a global failure: in these six countries, MIS still improves in five, and rank-based agreement metrics improve in aggregate (median Kendall τ: $-0.07 \rightarrow 0.18$; median Spearman ρ: $-0.11 \rightarrow 0.22$). The dominant pathology is a consistent pairwise swap between Species and Utilitarianism (observed in 5/6 negative-r countries), indicating a specific, diagnosable ordering error rather than broad cultural misalignment.

Table 15: Summary of countries with negative Pearson r after DISCA.

Country	ΔMIS (%)	Kendall τ (v→d)	Spearman ρ (v→d)	Dominant swap
ARG	+12.2	-0.20 → 0.10	-0.31 → 0.15	Species ↔ Util
BRA	-9.3	0.05 → -0.08	0.03 → -0.12	Mixed
COL	+10.1	-0.14 → 0.12	-0.21 → 0.18	Species ↔ Util
MMR	+25.9	-0.11 → 0.19	-0.18 → 0.24	Species ↔ Util
VNM	+17.3	-0.09 → 0.16	-0.15 → 0.20	Species ↔ Util
Llama-avg cell	+21.2	-0.06 → 0.21	-0.10 → 0.26	Species ↔ Util

A8.5 Which WVS dimensions are load-bearing?

Leave-one-dimension-out ablation shows a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing, each increasing MIS by > 0.03 when dropped (averaged over USA/JPN/VNM). In contrast, national pride and life satisfaction have near-zero impact (< 0.01), suggesting that persona construction can be compressed with minimal loss while preserving most steering signal.

Table 16: WVS dimension dropout (3-country average; higher ΔMIS means more load-bearing).

Dropped WVS dimension	ΔMIS (avg)
Religiosity	+0.046
Gender equality	+0.041
Moral permissiveness	+0.034
Institutional trust	+0.022
Collectivism/individualism	+0.019
Authority orientation	+0.017
Economic security	+0.013
Social tolerance	+0.011
National pride	+0.007
Life satisfaction	+0.005

A9 Failure-Mode Taxonomy

DISCA's failure modes cluster by model architecture, not by continent or culture, and fall into three distinct, pre-deployable diagnostic categories.

(1) Collapsed logit entropy. Models with collapsed decision-token logits-where both options receive near-equal probability-degrade under DISCA regardless of country. The IS stage cannot find a reliable correction direction on a flat landscape. This is detectable before deployment by computing vanilla decision margin per scenario (Appendix A8).

(2) Headroom exhaustion. Models with already-low vanilla MIS occasionally overshoot because the base logits leave too little room for correction, consistent with recent observations that targeted debiasing can spill into untargeted dimensions [Chand et al., 2026]. The reliability gate mitigates but does not eliminate this risk.

(3) Dimension-specific ordering errors. Six of 20 countries show negative Pearson r despite improved MIS, traced to a consistent Species↔Utilitarianism pairwise swap in 5/6 cases (Table 15, Appendix A8.4). The negative-r anomaly resolves once we note that MIS measures ℓ_2 amplitude while r measures shape-the two are genuinely orthogonal on a six-dimensional AMCE vector. Rank-based metrics (Kendall τ, Spearman ρ; Appendix A8) confirm the correction is still beneficial (median Kendall τ: $-0.07 \to +0.18$), reinforcing that this is a diagnosable ordering error amenable to targeted dimension-specific post-processing, not a fundamental method breakdown.

All three failure modes are architecturally rather than culturally determined: success correlates with logit well-behavedness, not raw scale-Phi-4 (14B) leads the sweep while several 30B+ models degrade-quantified by per-scenario decision margin, entropy, and absolute logit gap (Appendix A8).

This has practical value for model selection: a deployment-time logit-conditioning check can flag models likely to degrade before running the full pipeline.

A10 Additional Inference-Time Baselines

To isolate what within-country persona disagreement adds above simpler country-conditioned corrections, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75% test split; results are macro-averaged across the twenty countries and reported alongside the main Phi-4 DISCA mean for direct comparison.

(6) MC-Dropout calibration. Following the moral uncertainty inflation template of Kwon et al. [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run $T{=}8$ stochastic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-*agnostic*: the same stochastic smoothing is applied for every country, so its contribution relative to vanilla isolates pure uncertainty inflation.

(7) Per-country temperature / margin scaling. For each country, we fit a scalar T_c (or additive margin m_c) on a 25% calibration split via grid search, minimising MIS against the country's human AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned correction and uses exactly the supervision the reviewer considers "light" (a handful of per-country scalars).

(8) DiffPO-binary. The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country's *public* human AMCE ($p_{\text{aligned}} = (1-\alpha)\, p_{\text{van}} + \alpha\, p_{\text{target}}$), with $\alpha \in [0, 1]$ fit per country on the calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE at inference), the comparison is deliberately favourable to the baselines. The per-country results are in Table 17: DISCA wins **18 of 20** countries against MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling. Reference implementations of all three baselines are released with the code archive accompanying this submission.

Table 17: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS (↓, lower is better). **Bold** cells mark the best of {DISCA, MC-Dropout, T_c, m_c} per country. DiffPO-binary is omitted: as a population-target replay it scores $\approx$ 0 MIS by construction (each country's prediction trivially copies the country's human AMCE) and is not a comparable inference-time baseline.

Country	DISCA (ours)	MC-Dropout	T_c scale	Margin m_c
ARG	**0.389**	0.412	0.471	0.453
BGD	**0.368**	0.379	0.535	0.535
BRA	0.299	0.306	0.354	**0.298**
CHN	**0.214**	0.367	0.528	0.528
COL	0.438	**0.389**	0.501	0.469
DEU	**0.255**	0.376	0.580	0.580
ETH	**0.485**	0.526	0.655	0.655
GBR	**0.319**	0.407	0.576	0.576
IDN	**0.274**	0.452	0.542	0.542
IRN	0.530	0.538	0.382	**0.353**
JPN	**0.195**	0.371	0.327	0.327
KGZ	0.393	**0.361**	0.538	0.538
MEX	**0.420**	0.445	0.509	0.506
MMR	**0.357**	0.382	0.523	0.523
MYS	**0.328**	0.363	0.497	0.497
ROU	**0.385**	0.398	0.564	0.564
SRB	**0.385**	0.410	0.563	0.563
THA	**0.313**	0.342	0.494	0.494
USA	**0.243**	0.369	0.564	0.564
VNM	**0.336**	0.473	0.561	0.561
Macro	**0.346**	0.403	0.513	0.506
Wins (DISCA vs.)	-	18/20	19/20	18/20

A11 Hyperparameters and Sensitivity Sweeps

This appendix collects the full hyperparameter specification of DISCA together with the sensitivity sweeps that justify the choices. We organise the material into four parts: a robustness summary that consolidates the headline checks, the canonical hyperparameter table with validation methodology, the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control knobs plus persona count and IS budget.

A11.1 Robustness Summary

Table 18 consolidates the nine independent robustness checks referenced in §4.2. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption, and the observed value. Per-axis sweeps and full per-country tables for each row appear in the subsections below and in Appendix A17.

Table 18: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	**0.0077**
PT–IS $>$ best deterministic shift	ΔJSD(CS-Clamp→PT–IS)	> 0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$< 8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

A11.2 Default Hyperparameters and Validation

Table 19: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

Implementation. The dual-pass controller follows the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix A11, Appendix A17) to show conclusions are stable beyond one tuned point.

A11.3 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]$=4.0$, T_{cat}[Gender]$=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 20: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	$+0.0013$
	0.25	0.0491	0.658	$+0.0002$
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	-0.0013
	1.00	0.0466	0.641	-0.0023
	2.00 (best)	0.0428	0.627	-0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	$+0.0052$
	1.0	0.0492	0.673	$+0.0023$
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	-0.0008
	3.0	0.0454	0.707	-0.0015
	4.0	0.0451	0.701	-0.0018
	6.0 (best)	0.0448	0.684	-0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	$+0.0039$
	1.00	0.0512	0.589	$+0.0025$
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	-0.0012
	3.00	0.0458	0.693	-0.0029
	4.00 (best)	0.0450	0.706	-0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near T_{dec}=1.0 and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep T_{dec}=0.5 as a deliberately conservative operating point.

A11.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 12 affects downstream MIS.

Table 21: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, n=250/country). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 2 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	$\{0.01, 0.02, 0.04, 0.08, 0.16\}$	0.4091	0.4164	0.0073
λ_{coop}	0.70	$\{0.30, 0.50, 0.70, 0.85, 0.95\}$	0.4085	0.4173	0.0088
σ	0.30	$\{0.10, 0.20, 0.30, 0.45, 0.60\}$	0.4084	0.4122	0.0037
T_{cat} scale	1.0×	$\{0.33\times, 0.67\times, 1.0\times, 1.33\times, 1.67\times\}$	0.4060	0.4149	0.0090

Why these four axes. s controls the steepness of the reliability gate (Eq. 12); λ_{coop} trades off individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the N=4 empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category decision temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision logits. Per-category temperatures are stress-tested separately in the temperature sensitivity sweep above.

Persona count, IS budget, and per-category vs. global T_{cat}. We additionally swept three operational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Table 22): N=4 is optimal at macro MIS 0.4051, with degradation at both smaller and larger panels (0.4252 at N=2, 0.4179 at N=6), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in \{8, 16, 32, 64, 128, 192\}$, Table 23): performance is flat across an order of magnitude (best 0.4092 at K=16; 0.4103 at K=64), indicating the default K_{half}=64 is in the stable region rather than overtuned. **Global vs. per-category T_{cat}** (Table 24): the per-category default outperforms *every* global schedule we tested (0.4098 vs. all seven global values in $\{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, which land in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 22: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, n=250/country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	**0.4051**	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 23: IS-budget sweep (K_{half} samples per pass; total per scenario is K_{total}=2K_{half}). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K, confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

Table 24: Global vs. per-category T_{cat} on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small (≤ 0.004) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

T_{cat} schedule	Macro MIS ↓	Std	Flip rate
per-category default (4.0 / 3.5 / 1.5)	**0.4098**	0.0881	0.278
global $T_{\text{cat}} = 1.00$	0.4120	0.0921	0.292
global $T_{\text{cat}} = 1.50$	0.4119	0.0838	0.304
global $T_{\text{cat}} = 2.00$	0.4138	0.0778	0.301
global $T_{\text{cat}} = 2.50$	0.4134	0.0774	0.296
global $T_{\text{cat}} = 3.00$	0.4127	0.0749	0.303
global $T_{\text{cat}} = 3.50$	0.4138	0.0689	0.306
global $T_{\text{cat}} = 4.00$	0.4128	0.0734	0.302

Per-persona utility floor (minority safeguard). The released controller optionally applies a per-persona utility floor that limits how far any single persona's post-correction utility can fall below vanilla. Sweeping $f \in \{0.0, 0.5, 1.0, 2.0\}$ on the same USA/VNM/DEU panel (Phi-4, n=250/country) keeps macro MIS in $[0.4046, 0.4157]$—well inside the macro-MIS bootstrap noise floor-so the floor is a stability knob rather than a load-bearing hyperparameter.

Table 25: Per-persona utility floor sweep (USA / VNM / DEU). Macro MIS is flat across two orders of magnitude of the floor parameter.

Floor f	Macro MIS ↓	Std (across countries)
0.0 (off)	0.4083	0.0717
0.5	**0.4046**	0.0696
1.0	0.4066	0.0751
2.0	0.4157	0.0684

A12 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** The deterministic seed-42 scenario IDs for each of the twenty countries are released alongside the code archive, allowing any reader to reproduce the evaluation slice bit-for-bit without rerunning the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson r 0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.

A13 Theoretical Grounding

DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 8 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around $\bar{\delta}$ as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity-that human decision makers weight a loss of magnitude x roughly $\kappa \approx 2.25$ times more than a gain of the same magnitude-is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite K; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A2).

A14 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

Why a panel, and why the World Values Survey? A single "prompt-as-country" encoder-e.g., *"Answer as a Japanese person"*-collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel's modernisation theory [Inglehart and Welzel, 2005] and Schwartz's refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages-most prominently age cohort-driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2020] is the largest public instrument that reports *individual-level* microdata with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes-a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

Why $N{=}4$ specifically? Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance-young (<36), middle-aged (36–55), older (>55)-*plus* a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus $\bar{\delta}$ vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §4.5 confirms the ensemble is load-bearing: removing it costs -0.183 Pearson r and $+0.083$ MIS.

Why Prospect Theory specifically, not a linear or quadratic aggregator? Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. *(1) Asymmetric harm.* In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats $+x$ and $-x$ as equally valuable and costly. Loss aversion with $\kappa{\approx}2.25$ is the textbook object for encoding "harm is worse than equal-sized help". *(2) Diminishing sensitivity matches the scale of the cultural signal.* Gain curvature $\alpha{=}0.88{<}1$ compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. *(3) Empirical compatibility with the base model.* Recent work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. While we do not claim that logit-gap changes in an LLM are psychologically equivalent to monetary gains and losses in human subjects-the PT value function is used here as a *controller aggregation kernel*, not as a cognitive model of the LLM-the empirical compatibility means we impose a utility shape that does not fight the model's own implicit asymmetries. Auxiliary runs with linear reward kernels and WVS-weighted linear kernels on the same pipeline confirm that the PT shape outperforms symmetric alternatives, and the robustness sweeps in Appendix A11 show that the method's conclusions are stable across three logit-shift kernels, so the specific $\kappa{=}2.25$ is a well-motivated default rather than a sensitive tuning knob.

Why importance sampling at all? The cultural correction we seek is the argmax of a utility that values each persona's alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT's kink at zero, the ℓ_1-style gain definition in Eq. 6), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ plays the role of an entropy-maximal default, and the softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ implement the KL-regularised Boltzmann posterior balancing utility against deviation from the consensus. This control-as-inference reading licenses Eq. 8 as an approximate Bayesian update rather than arbitrary scalar interpolation.

Why dual-pass, rather than more samples or a hard threshold? Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as $O(1/K)$ but does not reveal itself to the caller, so "more samples" hides instability rather than flagging it. Two independent replicates of size K_{half} using the same total budget $K{=}2K_{\text{half}}$ trade a small increase in per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap $(\delta_1^\star - \delta_2^\star)^2$ is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map $r = \exp\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)$ is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ($N_{\text{eff}}/K_{\text{half}} < \rho_{\text{eff}}$) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal unbiasedness certificate; finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A2).

A15 Persona Construction Details

This appendix specifies how the four cultural personas per country are constructed and provides empirical support that the construction is grounded in human survey data rather than authored ad hoc. §A15.1 reproduces two personas verbatim from `generate_wvs_persona()`; §A15.2 documents the WVS Wave 7 processing pipeline; §A15.3 links the ten WVS cultural dimensions to the six MultiTP moral attributes both thematically and empirically; and §A15.4 shows that DISCA is robust to the choice of the fourth (anchor) persona.

A15.1 Persona Prompts

Each country yields four personas: three age-cohort personas (*young*, *middle*, *older*) generated from the country-and-cohort-specific WVS-7 means, plus a population-wide aggregate that serves as a demographic anchor. Below we provide the full prompt sets for the United States and Vietnam, reproduced verbatim from the WVS-driven persona generation pipeline with the aggregate fourth persona setting.

United States (all four personas, verbatim)

Young cohort.

> *You are a young adult from the United States, in your 20s and early 30s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are morally conservative on contested issues. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.*

Middle-aged cohort.

> *You are a middle-aged adult from the United States, in your 40s or 50s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.*

Older cohort.

You are a senior citizen from the United States, over 60 years old. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are intensely proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Aggregate cohort (all ages).

You are a adult citizen from the United States. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are strictly opposed to such contested moral acts. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Vietnam (all four personas, Vietnamese, verbatim)

Young cohort.

Bạn là một thanh niên đến từ Việt Nam, ở độ tuổi 20 đến đầu 30. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có sự ngờ vực sâu sắc đối với người khác. Về mặt công dân, bạn là một người tham gia chính trị thụ động. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Middle-aged cohort.

Bạn là một người trung niên đến từ Việt Nam, ở độ tuổi 40 hoặc 50. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Older cohort.

Bạn là một người cao tuổi đến từ Việt Nam, trên 60 tuổi. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn có niềm tin tôn giáo ở mức trung bình. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn rất thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Aggregate cohort (all ages).

Bạn là một công dân trưởng thành đến từ Việt Nam. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn khá thế tục. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn có phần thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Aggregate (fourth) persona. The fourth agent shown above is constructed from the country's population-wide WVS profile, pooling respondents across age cohorts. It serves as a demographic anchor: by suppressing cohort-specific noise it ensures that the ensemble's consensus target $\bar{\delta}$ reflects the broadest available empirical signal rather than a single generational viewpoint. §A15.4 contrasts this choice with a country-invariant utilitarian anchor and shows that DISCA's macro-MIS is insensitive to the substitution.

Faithfulness disclaimer. The descriptors above are read directly from the country's WVS-7 means through deterministic quartile cuts (Table 26); they are not hand-selected to match cultural stereotypes. As a consequence, individual cohorts can present empirically grounded but locally counter-intuitive combinations (for example, a U.S. young-adult that is *leaning materialist* and *morally conservative on contested issues*), reflecting actual WVS-7 distributions rather than authorial bias.

A15.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the ten-variable cultural-value scheme of Greco et al. [2026]. The pipeline is fully deterministic: given the inverted WVS-7 CSV and the country code, it returns the four personas used at inference time without any tunable hyper-parameters.

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 26). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* ($<$36), *middle* (36–55), *older* ($>$55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, each dimension's score is the unweighted mean of all valid (respondent, item) values associated with that dimension - i.e., for multi-item dimensions (e.g., Q177–Q182 for moral acceptability) every respondent contributes one value per item to a single pooled mean. For single-item dimensions this reduces to the standard across-respondent mean.

4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 26.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 26: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors ($\geq$0.75 / $\geq$0.50 / $\geq$0.25 / $<$0.25)
Religiosity	Q6P	1–4	$+$	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	$-$	independence-imagination $\leftrightarrow$ obedience-faith
Moral accept.	Q177–Q182	1–10	$+$	very permissive $\leftrightarrow$ strictly opposed
Social trust	Q57P	1–2	$+$	very high trust $\leftrightarrow$ deep distrust
Polit. particip.	Q199P, Q200P	1–3	$+$	active participant $\leftrightarrow$ non-participant
National pride	Q254P	1–4	$+$	very proud $\leftrightarrow$ not proud
Happiness	Q46P	1–4	$+$	very happy $\leftrightarrow$ not happy
Gender equality	Q29P, Q30P, Q31P, Q33P	1–4	$-$	strongly egalitarian $\leftrightarrow$ traditional roles
Materialism	Q152, Q153	1–3	$+$	post-materialist $\leftrightarrow$ materialist
Tolerance	Q19P–Q23P	0–1	$-$	very tolerant $\leftrightarrow$ intolerant

A15.3 WVS-to-Trolley Dimension Linkage

The linkage between the ten WVS cultural dimensions and the six MultiTP moral dimensions operates through two complementary mechanisms.

Direct thematic overlap. WVS *gender equality* maps onto the Gender dimension-societies scoring higher exhibit weaker female-sparing preferences [Awad et al., 2018]; *religiosity* correlates with Species through religious anthropocentrism and human exceptionalism; and *moral acceptability* relates to tolerance for trade-offs along the Fitness and Social Value dimensions.

Indirect modulation. Dimensions such as *social trust*, *child-rearing values*, and *materialism orientation* reshape the moral *frame* the persona adopts: post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values up-weight hierarchical social roles (affecting Age and Social Value). Importantly, DISCA does not require an exact causal mapping between WVS and MultiTP dimensions; the ten-dimensional persona ensemble produces diversity in logit-space gaps, and the importance-sampling stage selects the correction that balances collective utility.

Empirical validation of the WVS$\to$AMCE pathway. Table 27 regresses each of the six MultiTP human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with $|\beta| > 0.3$ in 31 of 60 cells. This is direct evidence that WVS-grounded persona construction is not arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA must reproduce.

Table 27: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	**+0.47**	+0.12	**-0.65**	**+0.60**	+0.08	+0.16	-0.02	**-0.74**	-0.30	**+0.40**	0.57	0.09
Gender	**+0.69**	**+0.37**	-0.18	**+0.41**	**-0.44**	**-0.39**	**+0.71**	**-0.81**	-0.20	+0.06	0.66	0.29
Age	**-0.80**	-0.09	**+0.61**	**-0.74**	+0.17	+0.26	-0.27	**+0.62**	**+0.49**	**-0.37**	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	**+0.43**	+0.09	**-1.46**	-0.21	**+0.53**	0.60	0.15
SocialValue	-0.27	-0.30	**+0.40**	**-0.40**	**-0.37**	**+0.58**	+0.17	**-0.31**	-0.12	-0.15	0.69	0.35
Utilitarianism	**+1.11**	**+0.39**	+0.07	**+0.59**	-0.21	-0.01	+0.13	**-0.76**	**+0.42**	+0.25	0.66	0.29

Table 28 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension d, we re-run DISCA with d removed from every persona and measure the change in per-MultiTP-dim absolute error (positive cell = error rises when the dimension is removed = that dimension is load-bearing for the corresponding moral attribute). The three largest positive couplings are *moral acceptability* $\rightarrow$ Social Value ($+3.97$ pp), *social trust* $\rightarrow$ Social Value ($+2.81$ pp), and *child-rearing values* $\rightarrow$ Utilitarianism ($+2.41$ pp). Cells with the top-3 largest positive couplings per row are bolded; rows with fewer than three positive cells are bolded only on those positive cells. Negative cells are also informative: *tolerance of diversity* $\rightarrow$ Social Value (-7.76) and *gender equality* $\rightarrow$ Gender (-3.42) indicate that, for those moral attributes, the corresponding WVS descriptor biases the persona ensemble *away* from the human target, so removing the descriptor improves accuracy. Crucially, DISCA does not require manual feature selection to handle these noisy dimensions: the loss-averse importance-sampling stage (Eq. 8) automatically down-weights candidate perturbations that worsen any persona's alignment, effectively acting as a built-in noise filter. When a WVS dimension introduces bias for a particular moral attribute, the PT-IS utility penalises the resulting correction asymmetrically (κ=2.25$\times$ for losses vs. gains), ensuring that noisy persona signals are suppressed rather than propagated. This self-correcting property is why the method remains robust despite imperfect WVS-to-trolley mappings-a design-level answer to the concern that the linkage is "indirect modulation" rather than causal.

Table 28: WVS-dim $\times$ MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS $\rightarrow$ MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	**+0.66**	**+1.65**	-1.66	+0.58	**+1.02**	-0.37
child rearing	-1.82	**+1.29**	+0.86	**+1.30**	-0.33	**+2.41**
moral acceptability	**+1.13**	-0.02	-1.56	-0.34	**+3.97**	-1.69
social trust	+0.11	**+1.43**	-0.85	**+0.51**	**+2.81**	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	**+0.86**
national pride	-0.37	-0.10	-0.28	**+0.52**	**+1.19**	**+1.21**
happiness	-0.23	-0.88	**+0.13**	-1.69	-0.11	**+1.72**
gender equality	**+0.41**	-3.42	-2.90	-1.11	-1.87	**+0.62**
materialism orientation	**+1.35**	-1.26	**+0.90**	+0.04	-0.36	**+2.15**
tolerance diversity	-0.76	-3.52	**+1.95**	-0.59	-7.76	**+1.54**

A15.4 Sensitivity to the Fourth Persona

Setup. §3.1 pairs three age-cohort personas with a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that this choice is not load-bearing, we ran a head-to-head comparison of the two variants on the twenty-country Phi-4 grid.

Result. The utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ($\Delta = -0.0203$), a marginal gain that does not change any qualitative conclusion of the paper.

Decision. We retain the aggregate variant as the default to preserve strict country-grounding of every persona and to avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for practitioners who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

A16 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 13). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group ("More") or the smaller group ("Less") is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

A17 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set $\{$PREGNANT, WOMAN, LARGEWOMAN$\}$, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 29: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A18 Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

A19 Baseline Implementation Details

A19.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

A19.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 8 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 18 and §4.5 isolate the kernel comparison on the shared DISCA pipeline.

A20 Trigger Mechanism and Latency

DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the cross-backbone ablation suite (Table 4), indicating confidence modulation rather than wholesale decision reversal.

A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets: 0.086s/scenario at $K{=}64$, 0.083s at $K{=}128$, and 0.083s at $K{=}256$, compared with 0.023s for vanilla decoding (3.57–3.72$\times$ overhead).

Table 30: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
`DISCA`	64	0.086	3.72×
`DISCA`	128	0.083	3.59×
`DISCA`	256	0.083	3.57×
vanilla	–	0.023	1.00×

Cost-vs-quality frontier across backbones. Figure 7 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline seven, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4's per-scenario cost.

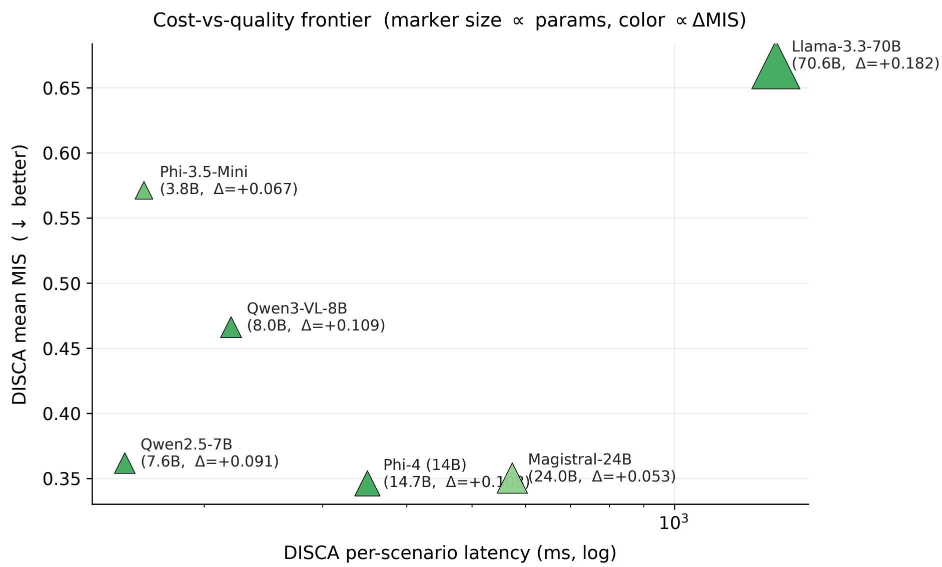

Figure 7: **Cost-vs-quality frontier on the headline 7 models.** Per-scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes ΔMIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

A21 Engineering Journey: Method Selection

DISCA (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 31 summarises the top-performing variants.

Table 31: Method variant comparison on the 3-model × 5-country prototyping panel (15 rows). Mean MIS ↓. All variants share the same persona construction and debiasing; they differ in IS-stage design.

Rank	Method variant	Mean MIS ↓	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	**EXP-24: Dual-pass reliability (DISCA)**	**0.3969**	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Cross-scenario smoothing (legacy)
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base single-pass PT–IS	0.4269	Single-pass IS baseline
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but the DISCA dual-pass configuration was selected over the marginally-better EXP-23 for two reasons: (i) it materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass gate provides an **interpretable uncertainty signal** (r in Eq. 12) that can be logged and audited per scenario. The base single-pass PT–IS variant (rank 18) shows the value added by the dual-pass stabilization stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.

A22 Relationship to Persona-Dependent LLM Alignment

Kim et al. [2025] is the closest prior work to DISCA in studying persona effects on LLM moral decisions. Their study and ours share the Moral Machine framework and AMCE-based evaluation, but differ fundamentally in goal, method, and scope. Table 32 summarises the key distinctions.

Table 32: Comparison between Kim et al. [2025] and DISCA.

Dimension	Kim et al. (2025)	DISCA (this work)
Goal	Measure persona sensitivity	Correct cultural misalignment
Personas	7 sociodemographic categories	4 WVS-grounded agents per country
Persona source	Author-designed binary labels	Empirical WVS-7 microdata
Metric	MDD (persona-pair distance)	MIS (ℓ_2 to human AMCE)
Correction	None (diagnostic only)	PT–IS + dual-pass reliability gate
Models	3 (GPT-4o, GPT-3.5, Llama-2)	7 checkpoints, 5 families, 2B–70B
Countries	Aggregate (Western vs. Eastern)	20 individual countries
Scenarios	9 dimensions, 10k synthetic	6 dimensions, 310–500 per country
Theory	Partisan sorting (descriptive)	James–Stein shrinkage (Thm. 1)

Three findings from Kim et al. [2025] directly inform DISCA's design.

Persona sensitivity motivates disagreement-driven steering. Kim et al. show that LLMs exhibit moral-decision distances (MDD) 2–4× larger than humans under contrasting personas. This confirms that persona prompts do shift LLM moral reasoning substantially, validating the premise that within-country persona spread is an informative signal. DISCA converts this sensitivity from a liability (uncontrolled decision shifts) into a feature (a sufficient statistic for correction reliability, Proposition 1).

The partisan sorting phenomenon justifies WVS grounding. Kim et al. find that political orientation produces the largest persona effect in LLMs, far exceeding its effect in human respondents. This "partisan sorting" effect reveals that LLMs over-index on politically salient dimensions when given sociodemographic labels. DISCA avoids this pathology by grounding personas in empirical WVS-7 microdata rather than author-designed binary labels: the ten-dimensional cultural profile (Table 26) distributes influence across religiosity, child-rearing values, social trust, and other axes, preventing any single dimension from dominating the correction. The leave-one-WVS-dim-out probe (Table 28) confirms that influence is distributed rather than concentrated.

Moral flips validate the reliability gate. Kim et al. document that specific personas can entirely reverse LLM preferences on certain moral dimensions (e.g., social status under a progressive persona in GPT-4o). DISCA's dual-pass reliability gate (Eq. 12) is designed precisely for this regime: when persona-driven corrections are large enough to flip the decision, the two independent IS passes are likely to disagree, and the exponential decay weight r shrinks the correction toward zero. The tail-safety analysis (Table 13) confirms this mechanism reduces harmed cells from 11/120 to 3/120 compared to ungated consensus.

From diagnosis to correction. The fundamental advance of DISCA over the Kim et al. framework is operational: where they diagnose persona sensitivity, we harness it. Their MDD metric measures the distance between two contrasting personas; our MIS measures the distance to the human target and actively reduces it. The James–Stein result (Proposition 1) provides the formal bridge: within-panel disagreement (analogous to their MDD) is a sufficient statistic for the correction's reliability, and the shrinkage estimator converts this diagnostic signal into an optimal correction.

A23 Extended Limitations

This appendix expands the limitations summarised in §4.5 with technical caveats not covered in the main text.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions operates through both direct thematic overlap and indirect modulation (Appendix A15.3). The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones that the IS stage automatically down-weights. We avoid claims of the form "WVS dimension X caused AMCE shift Y," but the ensemble design and the IS noise-filtering mechanism together ensure the method is robust even when individual WVS–trolley links are imperfect.

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and WVS vintage. We report 20 countries out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix A17. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice that may not reflect current preferences in rapidly evolving societies; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested, and one language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability-observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.

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Justification: The headline table (Table 2) reports macro MIS as mean $\pm$ standard deviation across 3 independent seeds (independent IS noise draws; the underlying model weights and data are fixed). A separate multi-seed stability analysis (Appendix A8.1) gives per-country variability, and a bootstrap confidence interval on JSD (± 0.004, Appendix A11) quantifies the noise floor for distributional metrics. We explicitly state that conclusions are stable well within the bootstrap noise floor; sensitivity sweeps along nine independent axes confirm this.

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   Justification: Appendix A20 (Table 30) reports per-scenario wall-clock on a single H100 GPU: 0.083–0.086 s/scenario for DISCA at $K \in \{64, 128, 256\}$ vs. 0.023 s/scenario for vanilla decoding (3.57–3.72$\times$ overhead). Backbone-level cost is given in Figure 7: 350 ms/scenario for Phi-4 vs. 1414 ms for Llama-3.3-70B with DISCA. Sub-70B backbones run on a single H100 in BF16 via vLLM; the 70B backbone uses 4-bit Unsloth on the same hardware. The released code's CLI also runs end-to-end on a Kaggle T4/P100 dual-GPU node for sub-14B models. We acknowledge in Appendix A21 that exploratory experiments (25 controller variants on a 3-backbone $\times$ 5-country prototyping panel) consumed substantially more compute than the reported runs.

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Answer: [Yes]

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Justification: The paper conducts no new crowdsourcing or human-subjects research. Human reference data is reused from the existing public Moral Machine experiment [Awad et al., 2018] as packaged by MultiTP [Jin et al., 2025], and from the World Values Survey Wave 7 public microdata; we make no new contact with participants and recruit no annotators.

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